

DAMA - DMBOK

DATA MANAGEMENT BODY OF KNOWLEDGE

2ND EDITION

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Dedicated to the memory of
Patricia Cupoli, MLS, MBA, CCP, CDMP

(May 25, 1948 – July 28, 2015)

for her lifelong commitment to the Data Management profession
and her contributions to this publication.

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Preface

DAMA International is pleased to release the second edition of the DAMA Guide to the Data Management Body of Knowledge (DAMA-DMBOK2). Since the publication of the first edition in 2009, significant developments have taken place in the field of data management. Data Governance has become a standard structure in many organizations, new technologies have enabled the collection and use of 'Big Data' (semi-structured and unstructured data in a wide range of formats), and the importance of data ethics has grown along with our ability to explore and exploit the vast amount of data and information produced as part of our daily lives.

These changes are exciting. They also place new and increasing demands on our profession. DAMA has responded to these changes by reformulating the DAMA Data Management Framework (the DAMA Wheel), adding detail and clarification, and expanding the scope of the DMBOK:

- Context diagrams for all Knowledge Areas have been improved and updated.
- Data Integration and Interoperability has been added as a new Knowledge Area to highlight its importance (Chapter 8).
- Data Ethics has been called out as a separate chapter due to the increasing necessity of an ethical approach to all aspects of data management (Chapter 2).
- The role of governance has been described both as a function (Chapter 3) and in relation to each Knowledge Area.
- A similar approach has been taken with organizational change management, which is described in Chapter 17 and incorporated into the Knowledge Area chapters.
- New chapters on Big Data and Data Science (Chapter 14) and Data Management Maturity Assessment (Chapter 15) help organizations understand where they want to go and give them the tools to get there.
- The second edition also includes a newly formulated set of data management principles to support the ability of organizations to manage their data effectively and get value from their data assets (Chapter 1).

We hope the DMBOK2 will serve data management professionals across the globe as a valuable resource and guide. Nevertheless, we also recognize it is only a starting point. Real advancement will come as we apply and learn from these ideas. DAMA exists to enable members to learn continuously, by sharing ideas, trends, problems, and solutions.

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Data Management

1. Introduction

Many organizations recognize that their data is a vital enterprise asset. Data and information can give them insight about their customers, products, and services. It can help them innovate and reach strategic goals. Despite that recognition, few organizations actively manage data as an asset from which they can derive ongoing value (Evans and Price, 2012). Deriving value from data does not happen in a vacuum or by accident. It requires intention, planning, coordination, and commitment. It requires management and leadership.

Data Management is the development, execution, and supervision of plans, policies, programs, and practices that deliver, control, protect, and enhance the value of data and information assets throughout their lifecycles.

A *Data Management Professional* is any person who works in any facet of data management (from technical management of data throughout its lifecycle to ensuring that data is properly utilized and leveraged) to meet strategic organizational goals. Data management professionals fill numerous roles, from the highly technical (e.g., database administrators, network administrators, programmers) to strategic business (e.g., Data Stewards, Data Strategists, Chief Data Officers).

Data management activities are wide-ranging. They include everything from the ability to make consistent decisions about how to get strategic value from data to the technical deployment and performance of databases. Thus data management requires both technical and non-technical (i.e., ‘business’) skills. Responsibility for managing data must be shared between business and information technology roles, and people in both areas must be able to collaborate to ensure an organization has high quality data that meets its strategic needs.

Data and information are not just assets in the sense that organizations invest in them in order to derive future value. Data and information are also vital to the day-to-day operations of most organizations. They have been called the ‘currency’, the ‘life blood’, and even the ‘new oil’ of the information economy.¹ Whether or not an organization gets value from its analytics, it cannot even transact business without data.

To support the data management professionals who carry out the work, DAMA International (The Data Management Association) has produced this book, the second edition of *The DAMA Guide to the Data*

¹ Google ‘data as currency’, ‘data as life blood’, and ‘the new oil’, for numerous references.

Management Body of Knowledge (DMBOK2). This edition builds on the first one, published in 2009, which provided foundational knowledge on which to build as the profession advanced and matured.

This chapter outlines a set of principles for data management. It discusses challenges related to following those principles and suggests approaches for meeting these challenges. The chapter also describes the DAMA Data Management Framework, which provides the context for the work carried out by data management professionals within various Data Management Knowledge Areas.

1.1 Business Drivers

Information and knowledge hold the key to competitive advantage. Organizations that have reliable, high quality data about their customers, products, services, and operations can make better decisions than those without data or with unreliable data. Failure to manage data is similar to failure to manage capital. It results in waste and lost opportunity. The primary driver for data management is to enable organizations to get value from their data assets, just as effective management of financial and physical assets enables organizations to get value from those assets.

1.2 Goals

Within an organization, data management goals include:

- Understanding and supporting the information needs of the enterprise and its stakeholders, including customers, employees, and business partners
- Capturing, storing, protecting, and ensuring the integrity of data assets
- Ensuring the quality of data and information
- Ensuring the privacy and confidentiality of stakeholder data
- Preventing unauthorized or inappropriate access, manipulation, or use of data and information
- Ensuring data can be used effectively to add value to the enterprise

2. Essential Concepts

2.1 Data

Long-standing definitions of *data* emphasize its role in representing facts about the world.² In relation to information technology, *data* is also understood as information that has been stored in digital form (though data is

² The New Oxford American Dictionary defines data as “facts and statistics collected together for analysis.” The American Society for Quality (ASQ) defines data as “A set of collected facts” and describes two kinds of numerical data: measured or variable and counted or attributed. The International Standards Organization (ISO) defines data as “re-interpretable

not limited to information that has been digitized and data management principles apply to data captured on paper as well as in databases). Still, because today we can capture so much information electronically, we call many things ‘data’ that would not have been called ‘data’ in earlier times – things like names, addresses, birthdates, what one ate for dinner on Saturday, the most recent book one purchased.

Such facts about individual people can be aggregated, analyzed, and used to make a profit, improve health, or influence public policy. Moreover our technological capacity to measure a wide range of events and activities (from the repercussions of the Big Bang to our own heartbeats) and to collect, store, and analyze electronic versions of things that were not previously thought of as data (videos, pictures, sound recordings, documents) is close to surpassing our ability to synthesize these data into usable information.³ To take advantage of the variety of data without being overwhelmed by its volume and velocity requires reliable, extensible data management practices.

Most people assume that, because data represents facts, it is a form of truth about the world and that the facts will fit together. But ‘facts’ are not always simple or straightforward. Data is a means of representation. It stands for things other than itself (Chisholm, 2010). Data is both an interpretation of the objects it represents and an object that must be interpreted (Sebastian-Coleman, 2013). This is another way of saying that we need context for data to be meaningful. Context can be thought of as data’s representational system; such a system includes a common vocabulary and a set of relationships between components. If we know the conventions of such a system, then we can interpret the data within it.⁴ These conventions are often documented in a specific kind of data referred to as Metadata.

However, because people often make different choices about how to represent concepts, they create different ways of representing the same concepts. From these choices, data takes on different shapes. Think of the range of ways we have to represent calendar dates, a concept about which there is an agreed-to definition. Now consider more complex concepts (such as customer or product), where the granularity and level of detail of what needs to be represented is not always self-evident, and the process of representation grows more complex, as does the process of managing that information over time. (See Chapter 10).

Even within a single organization, there are often multiple ways of representing the same idea. Hence the need for Data Architecture, modeling, governance, and stewardship, and Metadata and Data Quality management, all of which help people understand and use data. Across organizations, the problem of multiplicity multiplies. Hence the need for industry-level data standards that can bring more consistency to data.

Organizations have always needed to manage their data, but changes in technology have expanded the scope of this management need as they have changed people’s understanding of what data is. These changes have enabled organizations to use data in new ways to create products, share information, create knowledge, and improve

representation of information in a formalized manner suitable for communication, interpretation, or processing” (ISO 11179). This definition emphasizes the electronic nature of data and assumes, correctly, that data requires standards because it is managed through information technology systems. That said, it does not speak to the challenges of formalizing data in a consistent way, across disparate systems. Nor does it account well for the concept of unstructured data.

³ <http://ubm.io/2c4yPOJ> (Accessed 20016-12-04). <http://bit.ly/1rOQkt1> (Accessed 20016-12-04).

⁴ For additional information on the constructed-ness of data see: Kent, *Data and Reality* (2012) and Devlin, *Business Unintelligence* (2013).

organizational success. But the rapid growth of technology and with it human capacity to produce, capture, and mine data for meaning has intensified the need to manage data effectively.

2.2 Data and Information

Much ink has been spilled over the relationship between data and information. Data has been called the “raw material of information” and information has been called “data in context”.⁵ Often a layered pyramid is used to describe the relationship between data (at the base), information, knowledge, and wisdom (at the very top). While the pyramid can be helpful in describing why data needs to be well-managed, this representation presents several challenges for data management.

- It is based on the assumption that data simply exists. But data does not simply exist. Data has to be created.
- By describing a linear sequence from data through wisdom, it fails to recognize that it takes knowledge to create data in the first place.
- It implies that data and information are separate things, when in reality, the two concepts are intertwined with and dependent on each other. Data is a form of information and information is a form of data.

Within an organization, it may be helpful to draw a line between information and data for purposes of clear communication about the requirements and expectations of different uses by different stakeholders. (“Here is a sales report for the last quarter [information]. It is based on data from our data warehouse [data]. Next quarter these results [data] will be used to generate our quarter-over-quarter performance measures [information]”). Recognizing data and information need to be prepared for different purposes drives home a central tenet of data management: Both data and information need to be managed. Both will be of higher quality if they are managed together with uses and customer requirements in mind. Throughout the DMBOK, the terms will be used interchangeably.

2.3 Data as an Organizational Asset

An *asset* is an economic resource, that can be owned or controlled, and that holds or produces value. Assets can be converted to money. Data is widely recognized as an enterprise asset, though understanding of what it means to manage data as an asset is still evolving. In the early 1990s, some organizations found it questionable whether the value of goodwill should be given a monetary value. Now, the ‘value of goodwill’ commonly shows up as an item on the Profit and Loss Statement (P&L). Similarly, while not universally adopted, monetization of data is becoming increasingly common. It will not be too long before we see this as a feature of P&Ls. (See Chapter 3.)

Today’s organizations rely on their data assets to make more effective decisions and to operate more efficiently. Businesses use data to understand their customers, create new products and services, and improve operational efficiency by cutting costs and controlling risks. Government agencies, educational institutions, and not-for-profit

⁵ See English, 1999 and DAMA, 2009.

organizations also need high quality data to guide their operational, tactical, and strategic activities. As organizations increasingly depend on data, the value of data assets can be more clearly established.

Many organizations identify themselves as ‘data-driven’. Businesses aiming to stay competitive must stop making decisions based on gut feelings or instincts, and instead use event triggers and apply analytics to gain actionable insight. Being data-driven includes the recognition that data must be managed efficiently and with professional discipline, through a partnership of business leadership and technical expertise.

Furthermore, the pace of business today means that change is no longer optional; digital disruption is the norm. To react to this, business must co-create information solutions with technical data professionals working alongside line-of-business counterparts. They must plan for how to obtain and manage data that they know they need to support business strategy. They must also position themselves to take advantage of opportunities to leverage data in new ways.

2.4 Data Management Principles

Data management shares characteristics with other forms of asset management, as seen in Figure 1. It involves knowing what data an organization has and what might be accomplished with it, then determining how best to use data assets to reach organizational goals.

Like other management processes, it must balance strategic and operational needs. This balance can best be struck by following a set of principles that recognize salient features of data management and guide data management practice.

- **Data is an asset with unique properties:** Data is an asset, but it differs from other assets in important ways that influence how it is managed. The most obvious of these properties is that data is not consumed when it is used, as are financial and physical assets.
- **The value of data can and should be expressed in economic terms:** Calling data an asset implies that it has value. While there are techniques for measuring data’s qualitative and quantitative value, there are not yet standards for doing so. Organizations that want to make better decisions about their data should develop consistent ways to quantify that value. They should also measure both the costs of low quality data and the benefits of high quality data.
- **Managing data means managing the quality of data:** Ensuring that data is fit for purpose is a primary goal of data management. To manage quality, organizations must ensure they understand stakeholders’ requirements for quality and measure data against these requirements.
- **It takes Metadata to manage data:** Managing any asset requires having data about that asset (number of employees, accounting codes, etc.). The data used to manage and use data is called *Metadata*. Because data cannot be held or touched, to understand what it is and how to use it requires definition and knowledge in the form of Metadata. Metadata originates from a range of processes related to data creation, processing, and use, including architecture, modeling, stewardship, governance, Data Quality management, systems development, IT and business operations, and analytics.

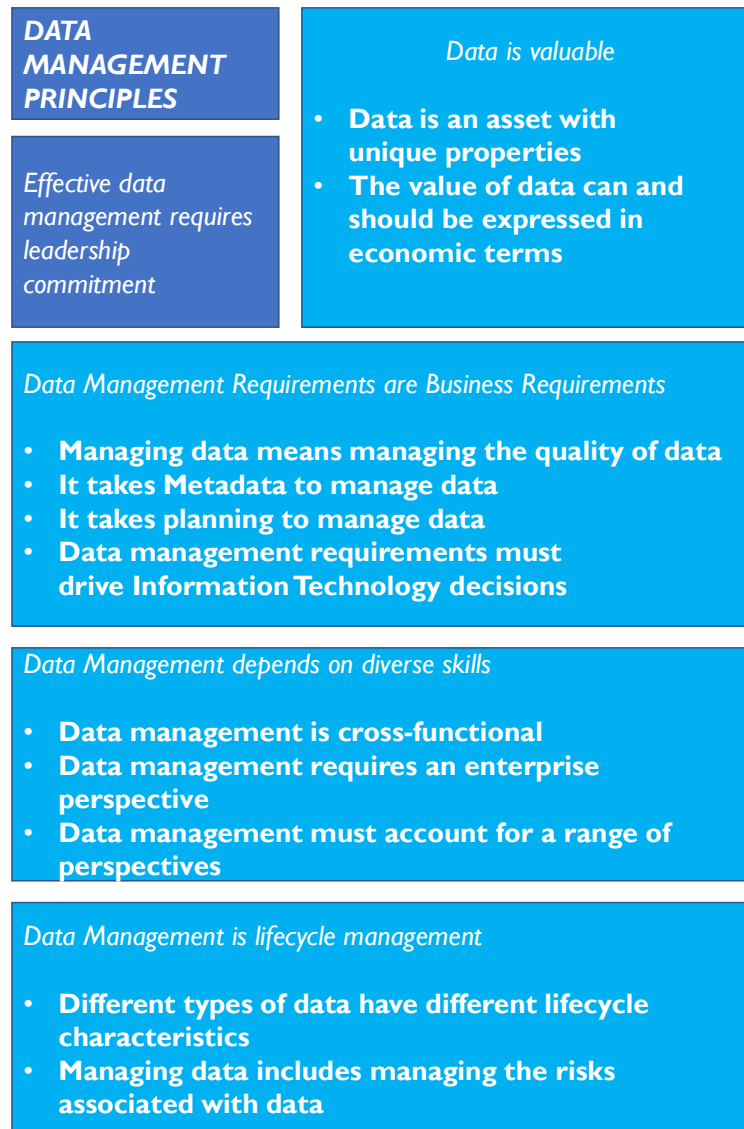


Figure 1 Data Management Principles

- **It takes planning to manage data:** Even small organizations can have complex technical and business process landscapes. Data is created in many places and is moved between places for use. To coordinate work and keep the end results aligned requires planning from an architectural and process perspective.
- **Data management is cross-functional; it requires a range of skills and expertise:** A single team cannot manage all of an organization's data. Data management requires both technical and non-technical skills and the ability to collaborate.
- **Data management requires an enterprise perspective:** Data management has local applications, but it must be applied across the enterprise to be as effective as possible. This is one reason why data management and data governance are intertwined.
- **Data management must account for a range of perspectives:** Data is fluid. Data management must constantly evolve to keep up with the ways data is created and used and the data consumers who use it.

- **Data management is lifecycle management:** Data has a lifecycle and managing data requires managing its lifecycle. Because data begets more data, the data lifecycle itself can be very complex. Data management practices need to account for the data lifecycle.
- **Different types of data have different lifecycle characteristics:** And for this reason, they have different management requirements. Data management practices have to recognize these differences and be flexible enough to meet different kinds of data lifecycle requirements.
- **Managing data includes managing the risks associated with data:** In addition to being an asset, data also represents risk to an organization. Data can be lost, stolen, or misused. Organizations must consider the ethical implications of their uses of data. Data-related risks must be managed as part of the data lifecycle.
- **Data management requirements must drive Information Technology decisions:** Data and data management are deeply intertwined with information technology and information technology management. Managing data requires an approach that ensures technology serves, rather than drives, an organization's strategic data needs.
- **Effective data management requires leadership commitment:** Data management involves a complex set of processes that, to be effective, require coordination, collaboration, and commitment. Getting there requires not only management skills, but also the vision and purpose that come from committed leadership.

2.5 Data Management Challenges

Because data management has distinct characteristics derived from the properties of data itself, it also presents challenges in following these principles. Details of these challenges are discussed in Sections 2.5.1 through 2.5.13. Many of these challenges refer to more than one principle.

2.5.1 Data Differs from Other Assets⁶

Physical assets can be pointed to, touched, and moved around. They can be in only one place at a time. Financial assets must be accounted for on a balance sheet. However, data is different. Data is not tangible. Yet it is durable; it does not wear out, though the value of data often changes as it ages. Data is easy to copy and transport. But it is not easy to reproduce if it is lost or destroyed. Because it is not consumed when used, it can even be stolen without being gone. Data is dynamic and can be used for multiple purposes. The same data can even be used by multiple people at the same time – something that is impossible with physical or financial assets. Many uses of data beget more data. Most organizations must manage increasing volumes of data and the relation between data sets.

⁶ This section derives from Redman, Thomas. *Data Quality for the Information Age* (1996) pp. 41-42, 232-36; and *Data Driven* (2008), Chapter One, "The Wondrous and Perilous Properties of Data and Information."

These differences make it challenging to put a monetary value on data. Without this monetary value, it is difficult to measure how data contributes to organizational success. These differences also raise other issues that affect data management, such as defining data ownership, inventorying how much data an organization has, protecting against the misuse of data, managing risk associated with data redundancy, and defining and enforcing standards for Data Quality.

Despite the challenges with measuring the value of data, most people recognize that data, indeed, has value. An organization's data is unique to itself. Were organizationally unique data (such as customer lists, product inventories, or claim history) to be lost or destroyed, replacing it would be impossible or extremely costly. Data is also the means by which an organization knows itself – it is a meta-asset that describes other assets. As such, it provides the foundation for organizational insight.

Within and between organizations, data and information are essential to conducting business. Most operational business transactions involve the exchange of information. Most information is exchanged electronically, creating a data trail. This data trail can serve purposes in addition to marking the exchanges that have taken place. It can provide information about how an organization functions.

Because of the important role that data plays in any organization, it needs to be managed with care.

2.5.2 Data Valuation

Value is the difference between the cost of a thing and the benefit derived from that thing. For some assets, like stock, calculating value is easy. It is the difference between what the stock cost when it was purchased and what it was sold for. But for data, these calculations are more complicated, because neither the costs nor the benefits of data are standardized.

Since each organization's data is unique to itself, an approach to data valuation needs to begin by articulating general cost and benefit categories that can be applied consistently within an organization. Sample categories include⁷:

- Cost of obtaining and storing data
- Cost of replacing data if it were lost
- Impact to the organization if data were missing
- Cost of risk mitigation and potential cost of risks associated with data
- Cost of improving data
- Benefits of higher quality data
- What competitors would pay for data
- What the data could be sold for
- Expected revenue from innovative uses of data

⁷ While the DMBOK2 was preparing to go to press, another means of valuing data was in the news: Wannacry ransomware attack (17 May 2017) impacted more than 100K organizations in 150 countries. The culprits used the software to hold data hostage until victims paid ransom to get their data released. <http://bit.ly/2tNoyQ7>.

A primary challenge to data asset valuation is that the value of data is contextual (what is of value to one organization may not be of value to another) and often temporal (what was valuable yesterday may not be valuable today). That said, within an organization, certain types of data are likely to be consistently valuable over time. Take reliable customer information, for example. Customer information may even grow more valuable over time, as more data accumulates related to customer activity.

In relation to data management, establishing ways to associate financial value with data is critical, since organizations need to understand assets in financial terms in order to make consistent decisions. Putting value on data becomes the basis of putting value on data management activities.⁸ The process of data valuation can also be used as a means of change management. Asking data management professionals and the stakeholders they support to understand the financial meaning of their work can help an organization transform its understanding of its own data and, through that, its approach to data management.

2.5.3 Data Quality

Ensuring that data is of high quality is central to data management. Organizations manage their data because they want to use it. If they cannot rely on it to meet business needs, then the effort to collect, store, secure, and enable access to it is wasted. To ensure data meets business needs, they must work with data consumers to define these needs, including characteristics that make data of high quality.

Largely because data has been associated so closely with information technology, managing Data Quality has historically been treated as an afterthought. IT teams are often dismissive of the data that the systems they create are supposed to store. It was probably a programmer who first observed ‘garbage in, garbage out’ – and who no doubt wanted to let it go at that. But the people who want to use the data cannot afford to be dismissive of quality. They generally assume data is reliable and trustworthy, until they have a reason to doubt these things. Once they lose trust, it is difficult to regain it.

Most uses of data involve learning from it in order to apply that learning and create value. Examples include understanding customer habits in order to improve a product or service and assessing organizational performance or market trends in order to develop a better business strategy, etc. Poor quality data will have a negative impact on these decisions.

As importantly, poor quality data is simply costly to any organization. Estimates differ, but experts think organizations spend between 10-30% of revenue handling data quality issues. IBM estimated the cost of poor quality data in the US in 2016 was \$3.1 Trillion.⁹ Many of the costs of poor quality data are hidden, indirect, and therefore hard to measure. Others, like fines, are direct and easy to calculate. Costs come from:

- Scrap and rework
- Work-arounds and hidden correction processes

⁸ For case studies and examples, see Aiken and Billings, *Monetizing Data Management* (2014).

⁹ Reported in Redman, Thomas. “Bad Data Costs U.S. \$3 Trillion per Year.” Harvard Business Review. 22 September 2016. <https://hbr.org/2016/09/bad-data-costs-the-u-s-3-trillion-per-year>.

- Organizational inefficiencies or low productivity
- Organizational conflict
- Low job satisfaction
- Customer dissatisfaction
- Opportunity costs, including inability to innovate
- Compliance costs or fines
- Reputational costs

The corresponding benefits of high quality data include:

- Improved customer experience
- Higher productivity
- Reduced risk
- Ability to act on opportunities
- Increased revenue
- Competitive advantage gained from insights on customers, products, processes, and opportunities

As these costs and benefits imply, managing Data Quality is not a one-time job. Producing high quality data requires planning, commitment, and a mindset that builds quality into processes and systems. All data management functions can influence Data Quality, for good or bad, so all of them must account for it as they execute their work. (See Chapter 13).

2.5.4 Planning for Better Data

As stated in the chapter introduction, deriving value from data does not happen by accident. It requires planning in many forms. It starts with the recognition that organizations can control how they obtain and create data. If they view data as a product that they create, they will make better decisions about it throughout its lifecycle. These decisions require systems thinking because they involve:

- The ways data connects business processes that might otherwise be seen as separate
- The relationship between business processes and the technology that supports them
- The design and architecture of systems and the data they produce and store
- The ways data might be used to advance organizational strategy

Planning for better data requires a strategic approach to architecture, modeling, and other design functions. It also depends on strategic collaboration between business and IT leadership. And, of course, it depends on the ability to execute effectively on individual projects.

The challenge is that there are usually organizational pressures, as well as the perennial pressures of time and money, that get in the way of better planning. Organizations must balance long- and short-term goals as they execute their strategies. Having clarity about the trade-offs leads to better decisions.

2.5.5 Metadata and Data Management

Organizations require reliable Metadata to manage data as an asset. Metadata in this sense should be understood comprehensively. It includes not only the business, technical, and operational Metadata described in Chapter 12, but also the Metadata embedded in Data Architecture, data models, data security requirements, data integration standards, and data operational processes. (See Chapters 4 – 11.)

Metadata describes what data an organization has, what it represents, how it is classified, where it came from, how it moves within the organization, how it evolves through use, who can and cannot use it, and whether it is of high quality. Data is abstract. Definitions and other descriptions of context enable it to be understood. They make data, the data lifecycle, and the complex systems that contain data comprehensible.

The challenge is that Metadata is a form of data and needs to be managed as such. Organizations that do not manage their data well generally do not manage their Metadata at all. Metadata management often provides a starting point for improvements in data management overall.

2.5.6 Data Management is Cross-functional

Data management is a complex process. Data is managed in different places within an organization by teams that have responsibility for different phases of the data lifecycle. Data management requires design skills to plan for systems, highly technical skills to administer hardware and build software, data analysis skills to understand issues and problems, analytic skills to interpret data, language skills to bring consensus to definitions and models, as well as strategic thinking to see opportunities to serve customers and meet goals.

The challenge is getting people with this range of skills and perspectives to recognize how the pieces fit together so that they collaborate well as they work toward common goals.

2.5.7 Establishing an Enterprise Perspective

Managing data requires understanding the scope and range of data within an organization. Data is one of the ‘horizontals’ of an organization. It moves across verticals, such as sales, marketing, and operations... Or at least it should. Data is not only unique to an organization; sometimes it is unique to a department or other sub-part of an organization. Because data is often viewed simply as a by-product of operational processes (for example, sales transaction records are the by-product of the selling process), it is not always planned for beyond the immediate need.

Even within an organization, data can be disparate. Data originates in multiple places within an organization. Different departments may have different ways of representing the same concept (e.g., customer, product, vendor). As anyone involved in a data integration or Master Data Management project can testify, subtle (or blatant) differences in representational choices present challenges in managing data across an organization. At the same time, stakeholders assume that an organization’s data should be coherent, and a goal of managing data is to make it fit together in common sense ways so that it is usable by a wide range of data consumers.

One reason data governance has become increasingly important is to help organizations make decisions about data across verticals. (See Chapter 3.)

2.5.8 Accounting for Other Perspectives

Today's organizations use data that they create internally, as well as data that they acquire from external sources. They have to account for different legal and compliance requirements across national and industry lines. People who create data often forget that someone else will use that data later. Knowledge of the potential uses of data enables better planning for the data lifecycle and, with that, for better quality data. Data can also be misused. Accounting for this risk reduces the likelihood of misuse.

2.5.9 The Data Lifecycle

Like other assets, data has a lifecycle. To effectively manage data assets, organizations need to understand and plan for the data lifecycle. Well-managed data is managed strategically, with a vision of how the organization will use its data. A strategic organization will define not only its data content requirements, but also its data management requirements. These include policies and expectations for use, quality, controls, and security; an enterprise approach to architecture and design; and a sustainable approach to both infrastructure and software development.

The data lifecycle is based on the product lifecycle. It should not be confused with the systems development lifecycle. Conceptually, the data lifecycle is easy to describe (see Figure 2). It includes processes that create or obtain data, those that move, transform, and store it and enable it to be maintained and shared, and those that use or apply it, as well as those that dispose of it.¹⁰ Throughout its lifecycle, data may be cleansed, transformed, merged, enhanced, or aggregated. As data is used or enhanced, new data is often created, so the lifecycle has internal iterations that are not shown on the diagram. Data is rarely static. Managing data involves a set of interconnected processes aligned with the data lifecycle.

The specifics of the data lifecycle within a given organization can be quite complicated, because data not only has a lifecycle, it also has lineage (i.e., a pathway along which it moves from its point of origin to its point of usage, sometimes called the *data chain*). Understanding the data lineage requires documenting the origin of data sets, as well as their movement and transformation through systems where they are accessed and used. Lifecycle and lineage intersect and can be understood in relation to each other. The better an organization understands the lifecycle and lineage of its data, the better able it will be to manage its data.

The focus of data management on the data lifecycle has several important implications:

- **Creation and usage are the most critical points in the data lifecycle:** Data management must be executed with an understanding of how data is produced, or obtained, as well as how data is used. It costs money to produce data. Data is valuable only when it is consumed or applied. (See Chapters 5, 6, 8, 11, and 14.)

¹⁰ See McGilvray (2008) and English (1999) for information on the product lifecycle and data.

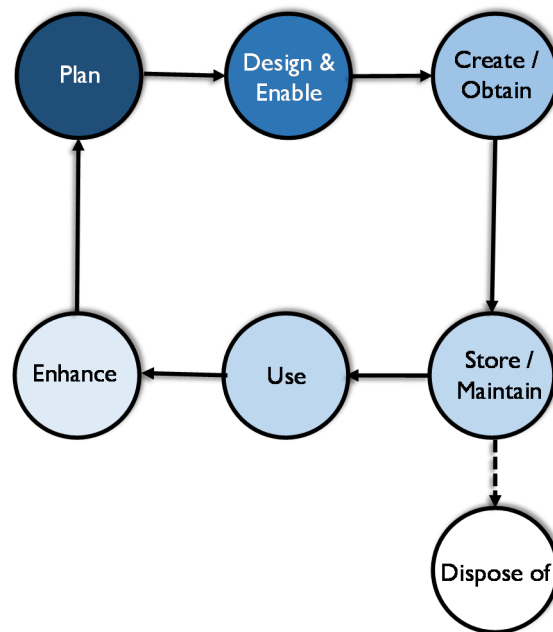


Figure 2 Data Lifecycle Key Activities

- **Data Quality must be managed throughout the data lifecycle:** Data Quality Management is central to data management. Low quality data represents cost and risk, rather than value. Organizations often find it challenging to manage the quality of data because, as described previously, data is often created as a by-product or operational processes and organizations often do not set explicit standards for quality. Because the quality of quality can be impacted by a range of lifecycle events, quality must be planned for as part of the data lifecycle (see Chapter 13).
- **Metadata Quality must be managed through the data lifecycle:** Because Metadata is a form of data, and because organizations rely on it to manage other data, Metadata quality must be managed in the same way as the quality of other data (see Chapter 12).
- **Data Security must be managed throughout the data lifecycle:** Data management also includes ensuring that data is secure and that risks associated with data are mitigated. Data that requires protection must be protected throughout its lifecycle, from creation to disposal (see Chapter 7 Data Security).
- **Data Management efforts should focus on the most critical data:** Organizations produce a lot of data, a large portion of which is never actually used. Trying to manage every piece of data is not possible. Lifecycle management requires focusing on an organization's most critical data and minimizing data ROT (Data that is Redundant, Obsolete, Trivial) (Aiken, 2014).

2.5.10 Different Types of Data

Managing data is made more complicated by the fact that there are different types of data that have different lifecycle management requirements. Any management system needs to classify the objects that are managed. Data

can be classified by type of data (e.g., transactional data, Reference Data, Master Data, Metadata; alternatively category data, resource data, event data, detailed transaction data) or by content (e.g., data domains, subject areas) or by format or by the level of protection the data requires. Data can also be classified by how and where it is stored or accessed. (See Chapters 5 and 10.)

Because different types of data have different requirements, are associated with different risks, and play different roles within an organization, many of the tools of data management are focused on aspects of classification and control (Bryce, 2005). For example, Master Data has different uses and consequently different management requirements than does transactional data. (See Chapters 9, 10, 12, and 14.)

2.5.11 Data and Risk

Data not only represents value, it also represents risk. Low quality data (inaccurate, incomplete, or out-of-date) obviously represents risk because its information is not right. But data is also risky because it can be misunderstood and misused.

Organizations get the most value from the highest quality data – available, relevant, complete, accurate, consistent, timely, usable, meaningful, and understood. Yet, for many important decisions, we have information gaps – the difference between what we know and what we need to know to make an effective decision. Information gaps represent enterprise liabilities with potentially profound impacts on operational effectiveness and profitability. Organizations that recognize the value of high quality data can take concrete, proactive steps to improve the quality and usability of data and information within regulatory and ethical cultural frameworks.

The increased role of information as an organizational asset across all sectors has led to an increased focus by regulators and legislators on the potential uses and abuses of information. From Sarbanes-Oxley (focusing on controls over accuracy and validity of financial transaction data from transaction to balance sheet) to Solvency II (focusing on data lineage and quality of data underpinning risk models and capital adequacy in the insurance sector), to the rapid growth in the last decade of data privacy regulations (covering the processing of data about people across a wide range of industries and jurisdictions), it is clear that, while we are still waiting for Accounting to put Information on the balance sheet as an asset, the regulatory environment increasingly expects to see it on the risk register, with appropriate mitigations and controls being applied.

Likewise, as consumers become more aware of how their data is used, they expect not only smoother and more efficient operation of processes, but also protection of their information and respect for their privacy. This means the scope of who our strategic stakeholders are as data management professionals can often be broader than might have traditionally been the case. (See Chapters 2 Data Handling Ethics and 7 Data Security.)

Increasingly, the balance sheet impact of information management, unfortunately, all too often arises when these risks are not managed and shareholders vote with their share portfolios, regulators impose fines or restrictions on operations, and customers vote with their wallets.

2.5.12 Data Management and Technology

As noted in the chapter introduction and elsewhere, data management activities are wide-ranging and require both technical and business skills. Because almost all of today's data is stored electronically, data management tactics are strongly influenced by technology. From its inception, the concept of data management has been deeply intertwined with management of technology. That legacy continues. In many organizations, there is ongoing tension between the drive to build new technology and the desire to have more reliable data – as if the two were opposed to each other instead of necessary to each other.

Successful data management requires sound decisions about technology, but managing technology is not the same as managing data. Organizations need to understand the impact of technology on data, in order to prevent technological temptation from driving their decisions about data. Instead, data requirements aligned with business strategy should drive decisions about technology.

2.5.13 Effective Data Management Requires Leadership and Commitment

The Leader's Data Manifesto (2017) recognized that an “organization's best opportunities for organic growth lie in data.” Although most organizations recognize their data as an asset, they are far from being data-driven. Many don't know what data they have or what data is most critical to their business. They confuse data and information technology and mismanage both. They do not approach data strategically. And they underestimate the work involved with data management. These conditions add to the challenges of managing data and point to a factor critical to an organization's potential for success: committed leadership and the involvement of everyone at all levels of the organization.¹¹

The challenges outlined here should drive this point home: Data management is neither easy nor simple. But because few organizations do it well, it is a source of largely untapped opportunity. To become better at it requires vision, planning, and willingness to change. (See Chapters 15-17.)

Advocacy for the role of Chief Data Officer (CDO) stems from a recognition that managing data presents unique challenges and that successful data management must be business-driven, rather than IT-driven. A CDO can lead data management initiatives and enable an organization to leverage its data assets and gain competitive advantage from them. However, a CDO not only leads initiatives. He or she must also lead cultural change that enables an organization to have a more strategic approach to its data.

2.6 Data Management Strategy

A strategy is a set of choices and decisions that together chart a high-level course of action to achieve high-level goals. In the game of chess, a strategy is a sequenced set of moves to win by checkmate or to survive by stalemate. A strategic *plan* is a high-level course of action to achieve high-level goals.

¹¹ The full text of *The Leader's Data Manifesto* can be found at: <http://bit.ly/2sQhcy7>.

A data strategy should include business plans to use information to competitive advantage and support enterprise goals. Data strategy must come from an understanding of the data needs inherent in the business strategy: what data the organization needs, how it will get the data, how it will manage it and ensure its reliability over time, and how it will utilize it.

Typically, a data strategy requires a supporting Data Management program strategy – a plan for maintaining and improving the quality of data, data integrity, access, and security while mitigating known and implied risks. The strategy must also address known challenges related to data management.

In many organizations, the data management strategy is owned and maintained by the CDO and enacted through a data governance team, supported by a Data Governance Council. Often, the CDO will draft an initial data strategy and data management strategy even before a Data Governance Council is formed, in order to gain senior management's commitment to establishing data stewardship and governance.

The components of a data management strategy should include:

- A compelling vision for data management
- A summary business case for data management, with selected examples
- Guiding principles, values, and management perspectives
- The mission and long-term directional goals of data management
- Proposed measures of data management success
- Short-term (12-24 months) Data Management program objectives that are SMART (specific, measurable, actionable, realistic, time-bound)
- Descriptions of data management roles and organizations, along with a summary of their responsibilities and decision rights
- Descriptions of Data Management program components and initiatives
- A prioritized program of work with scope boundaries
- A draft implementation roadmap with projects and action items

Deliverables from strategic planning for data management include:

- **A Data Management Charter:** Overall vision, business case, goals, guiding principles, measures of success, critical success factors, recognized risks, operating model, etc.
- **A Data Management Scope Statement:** Goals and objectives for some planning horizon (usually 3 years) and the roles, organizations, and individual leaders accountable for achieving these objectives.
- **A Data Management Implementation Roadmap:** Identifying specific programs, projects, task assignments, and delivery milestones (see Chapter 15).

The data management strategy should address all DAMA Data Management Framework Knowledge Areas relevant to the organization. (See Figure 5 Figure 5 The DAMA-DMBOK2 Data Management Framework (The DAMA Wheel and Sections 3.3 and 4.)

3. Data Management Frameworks

Data management involves a set of interdependent functions, each with its own goals, activities, and responsibilities. Data management professionals need to account for the challenges inherent in trying to derive value from an abstract enterprise asset while balancing strategic and operational goals, specific business and technical requirements, risk and compliance demands, and conflicting understandings of what the data represents and whether it is of high quality.

There is a lot to keep track of, which is why it helps to have a framework to understand the data management comprehensively and see relationships between its component pieces. Because the functions depend on one another and need to be aligned, in any organization, people responsible for the different aspects of data management need to collaborate if the organization is to derive value from its data.

Frameworks developed at different levels of abstraction provide a range of perspectives on how to approach data management. These perspectives provide insight that can be used to clarify strategy, develop roadmaps, organize teams, and align functions.

The ideas and concepts presented in the DMBOK2 will be applied differently across organizations. An organization's approach to data management depends on key factors such as its industry, the range of data it uses, its culture, maturity level, strategy, vision, and the specific challenges it is addressing. The frameworks described in this section provide some lenses through which to see data management and apply concepts presented in the DMBOK.

- The first two, the Strategic Alignment Model and the Amsterdam Information Model show high-level relationships that influence how an organization manages data.
- The DAMA DMBOK Framework (The DAMA Wheel, Hexagon, and Context Diagram) describes Data Management Knowledge Areas, as defined by DAMA, and explains how their visual representation within the DMBOK.
- The final two take the DAMA Wheel as a starting point and rearrange the pieces in order to better understand and describe the relationships between them.

3.1 Strategic Alignment Model

The Strategic Alignment Model (Henderson and Venkatraman, 1999) abstracts the fundamental drivers for any approach to data management. At its center is the relationship between data and information. Information is most often associated with business strategy and the operational use of data. Data is associated with information

technology and processes which support physical management of systems that make data accessible for use. Surrounding this concept are the four fundamental domains of strategic choice: business strategy, information technology strategy, organizational infrastructure and processes, and information technology infrastructure and processes.

The fully articulated Strategic Alignment Model is more complex than is illustrated in Figure 3. Each of the corner hexagons has its own underlying dimensions. For example, within both Business and IT strategy, there is a need to account for scope, competencies, and governance. Operations must account for infrastructure, processes, and skills. The relationships between the pieces help an organization understand both the strategic fit of the different components and the functional integration of the pieces. Even the high-level depiction of the model is useful in understanding the organizational factors that influence decisions about data and data management.

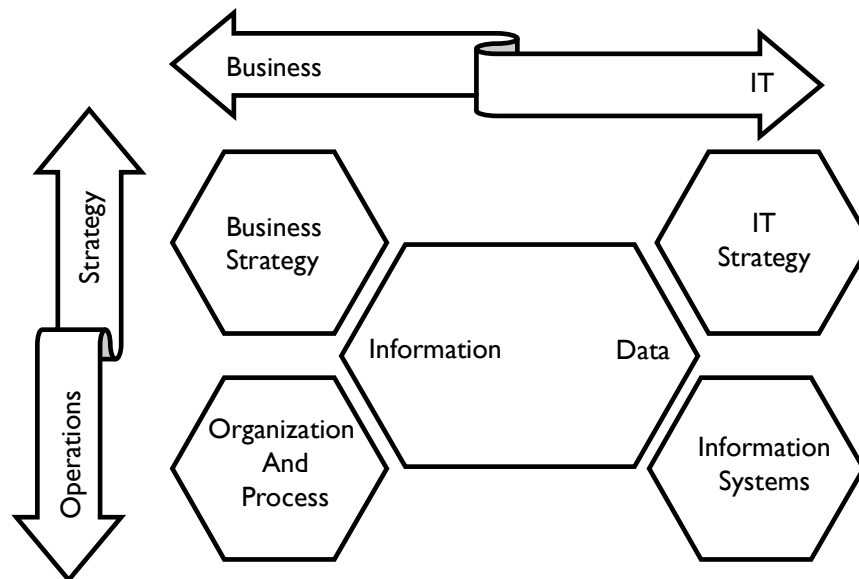


Figure 3 Strategic Alignment Model¹²

3.2 The Amsterdam Information Model

The Amsterdam Information Model, like the Strategic Alignment Model, takes a strategic perspective on business and IT alignment (Abcouwer, Maes, and Truijens, 1997),¹³ Known as the 9-cell, it recognizes a middle layer that focuses on structure and tactics, including planning and architecture. Moreover, it recognizes the necessity of information communication (expressed as the information governance and data quality pillar in Figure 4).

The creators of both the SAM and AIM frameworks describe in detail the relation between the components, from both a horizontal (Business / IT strategy) and vertical (Business Strategy / Business Operations) perspective.

¹² Adapted by Henderson and Venkatraman.

¹³ See also: Business IT Alignment Blog, *The Amsterdam Information Model (AIM) 9-Cells* (posted 2010-12-08). <https://businessitalignment.wordpress.com/tag/amsterdam-information-model/> *Frameworks for IT Management*, Chapter 13. Van Haren Publishing, 2006. <http://bit.ly/2sq2Ow1>.

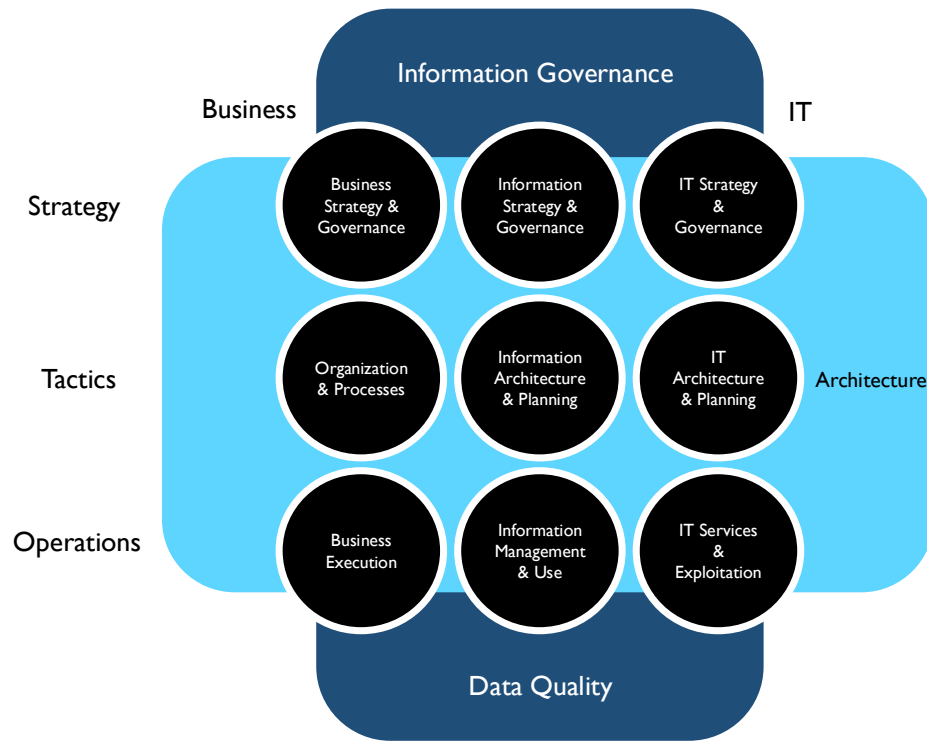


Figure 4 Amsterdam Information Model¹⁴

3.3 The DAMA-DMBOK Framework

The DAMA-DMBOK Framework goes into more depth about the Knowledge Areas that make up the overall scope of data management. Three visuals depict DAMA's Data Management Framework:

- The DAMA Wheel (Figure 5)
- The Environmental Factors hexagon (Figure 6)
- The Knowledge Area Context Diagram (Figure 7)

The DAMA Wheel defines the Data Management Knowledge Areas. It places data governance at the center of data management activities, since governance is required for consistency within and balance between the functions. The other Knowledge Areas (Data Architecture, Data Modeling, etc.) are balanced around the Wheel. They are all necessary parts of a mature data management function, but they may be implemented at different times, depending on the requirements of the organization. These Knowledge Areas are the focus of Chapters 3 – 13 of the DMBOK2. (See Figure 5.)

The Environmental Factors hexagon shows the relationship between people, process, and technology and provides a key for reading the DMBOK context diagrams. It puts goals and principles at the center, since these provide guidance for how people should execute activities and effectively use the tools required for successful data management. (See Figure 6.)

¹⁴ Adapted from Maas.

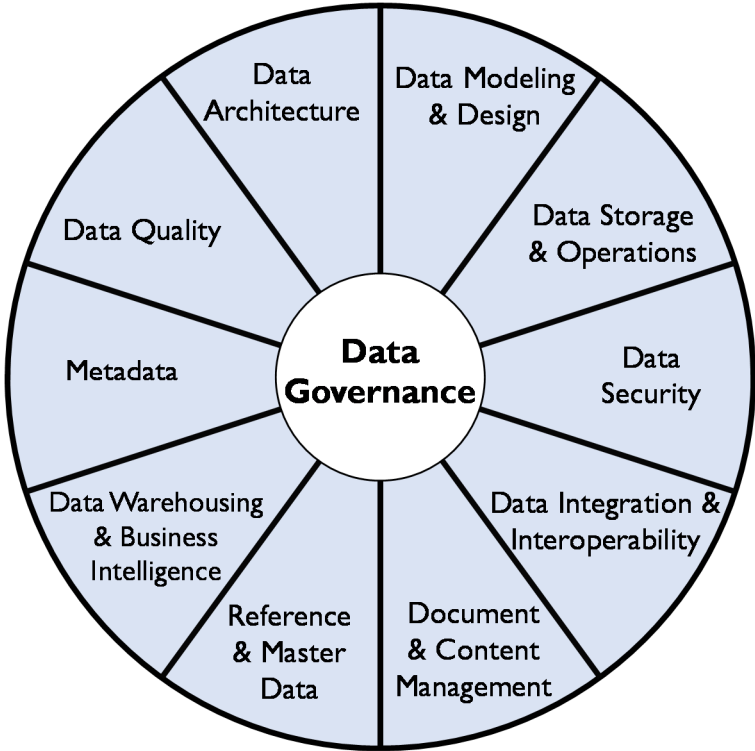


Figure 5 The DAMA-DMBOK2 Data Management Framework (The DAMA Wheel)



Figure 6 DAMA Environmental Factors Hexagon

The Knowledge Area Context Diagrams (See Figure 7) describe the detail of the Knowledge Areas, including detail related to people, processes and technology. They are based on the concept of a SIPOC diagram used for product

management (Suppliers, Inputs, Processes, Outputs, and Consumers). Context Diagrams put activities at the center, since they produce the deliverables that meet the requirements of stakeholders.

Each context diagram begins with the Knowledge Area's definition and goals. Activities that drive the goals (center) are classified into four phases: Plan (P), Develop (D), Operate (O), and Control (C). On the left side (flowing into the activities) are the Inputs and Suppliers. On the right side (flowing out of the activities) are Deliverables and Consumers. Participants are listed below the Activities. On the bottom are Tools, Techniques, and Metrics that influence aspects of the Knowledge Area.

Lists in the context diagram are illustrative, not exhaustive. Items will apply differently to different organizations. The high-level role lists include only the most important roles. Each organization can adapt this pattern to address its own needs.

GENERIC CONTEXT DIAGRAM

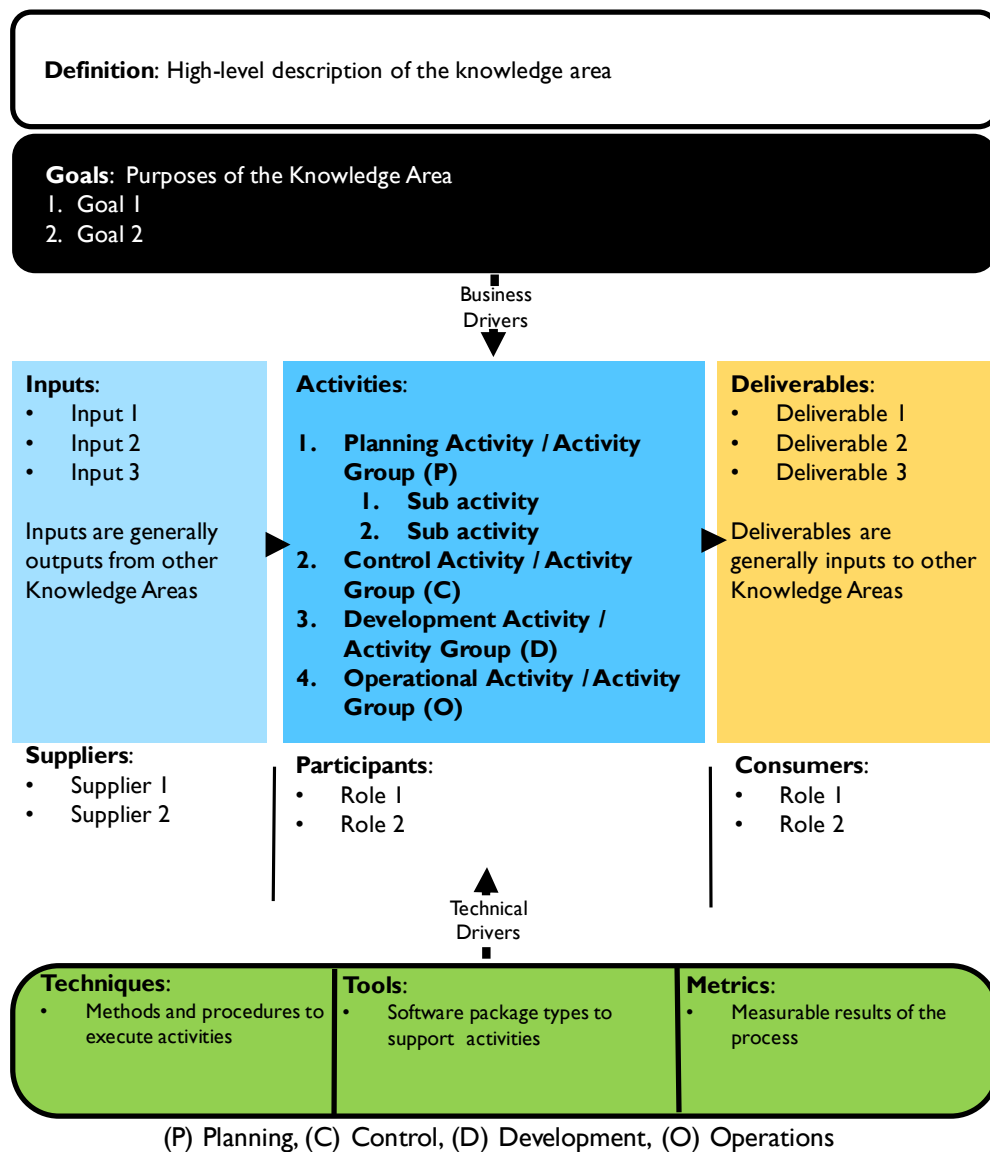


Figure 7 Knowledge Area Context Diagram

The component pieces of the context diagram include:

1. **Definition:** This section concisely defines the Knowledge Area.
2. **Goals** describe the purpose the Knowledge Area and the fundamental principles that guide performance of activities within each Knowledge Area.
3. **Activities** are the actions and tasks required to meet the goals of the Knowledge Area. Some activities are described in terms of sub-activities, tasks, and steps. Activities are classified into four categories: Plan, Develop, Operate, and Control.
 - a. **(P) Planning Activities** set the strategic and tactical course for meeting data management goals. Planning activities occur on a recurring basis.
 - b. **(D) Development Activities** are organized around the system development lifecycle (SDLC) (analysis, design, build, test, preparation, and deployment).
 - c. **(C) Control Activities** ensure the ongoing quality of data and the integrity, reliability, and security of systems through which data is accessed and used.
 - d. **(O) Operational Activities** support the use, maintenance, and enhancement of systems and processes through which data is accessed and used.
4. **Inputs** are the tangible things that each Knowledge Area requires to initiate its activities. Many activities require the same inputs. For example, many require knowledge of the Business Strategy as input.
5. **Deliverables** are the outputs of the activities within the Knowledge Area, the tangible things that each function is responsible for producing. Deliverables may be ends in themselves or inputs into other activities. Several primary deliverables are created by multiple functions.
6. **Roles and Responsibilities** describe how individuals and teams contribute to activities within the Knowledge Area. Roles are described conceptually, with a focus on groups of roles required in most organizations. Roles for individuals are defined in terms of skills and qualification requirements. Skills Framework for the Information Age (SFIA) was used to help align role titles. Many roles will be cross-functional.¹⁵ (See Chapter 16).
7. **Suppliers** are the people responsible for providing or enabling access to inputs for the activities.
8. **Consumers** those that directly benefit from the primary deliverables created by the data management activities.
9. **Participants** are the people that perform, manage the performance of, or approve the activities in the Knowledge Area.

¹⁵ <http://bit.ly/2sTusD0>.

10. **Tools** are the applications and other technologies that enable the goals of the Knowledge Area.¹⁶
11. **Techniques** are the methods and procedures used to perform activities and produce deliverables within a Knowledge Area. Techniques include common conventions, best practice recommendations, standards and protocols, and, where applicable, emerging alternative approaches.
12. **Metrics** are standards for measurement or evaluation of performance, progress, quality, efficiency, or other effect. The metrics sections identify measurable facets of the work that is done within each Knowledge Area. Metrics may also measure more abstract characteristics, like improvement or value.

While the DAMA Wheel presents the set of Knowledge Areas at a high level, the Hexagon recognizes components of the structure of Knowledge Areas, and the Context Diagrams present the detail within each Knowledge Area. None of the pieces of the existing DAMA Data Management framework describe the relationship between the different Knowledge Areas. Efforts to address that question have resulted in reformulations of the DAMA Framework, which are described in the next two sections.

3.4 DMBOK Pyramid (Aiken)

If asked, many organizations would say that they want to get the most out of their data – they are striving for that golden pyramid of advanced practices (data mining, analytics, etc.). But that pyramid is only the top of a larger structure, a pinnacle on a foundation. Most organizations do not have the luxury of defining a data management strategy before they start having to manage data. Instead, they build toward that capability, most times under less than optimal conditions.

Peter Aiken’s framework uses the DMBOK functional areas to describe the situation in which many organizations find themselves. An organization can use it to define a way forward to a state where they have reliable data and processes to support strategic business goals. In trying to reach this goal, many organizations go through a similar logical progression of steps (See Figure 8):

- **Phase 1:** The organization purchases an application that includes database capabilities. This means the organization has a starting point for data modeling / design, data storage, and data security (e.g., let some people in and keep others out). To get the system functioning within their environment and with their data requires work on integration and interoperability.
- **Phase 2:** Once they start using the application, they will find challenges with the quality of their data. But getting to higher quality data depends on reliable Metadata and consistent Data Architecture. These provide clarity on how data from different systems works together.
- **Phase 3:** Disciplined practices for managing Data Quality, Metadata, and architecture require Data Governance that provides structural support for data management activities. Data Governance also enables execution of strategic initiatives, such as Document and Content Management, Reference Data

¹⁶ DAMA International does not endorse specific tools or vendors.

Management, Master Data Management, Data Warehousing, and Business Intelligence, which fully enable the advanced practices within the golden pyramid.

- **Phase 4:** The organization leverages the benefits of well-managed data and advances its analytic capabilities.

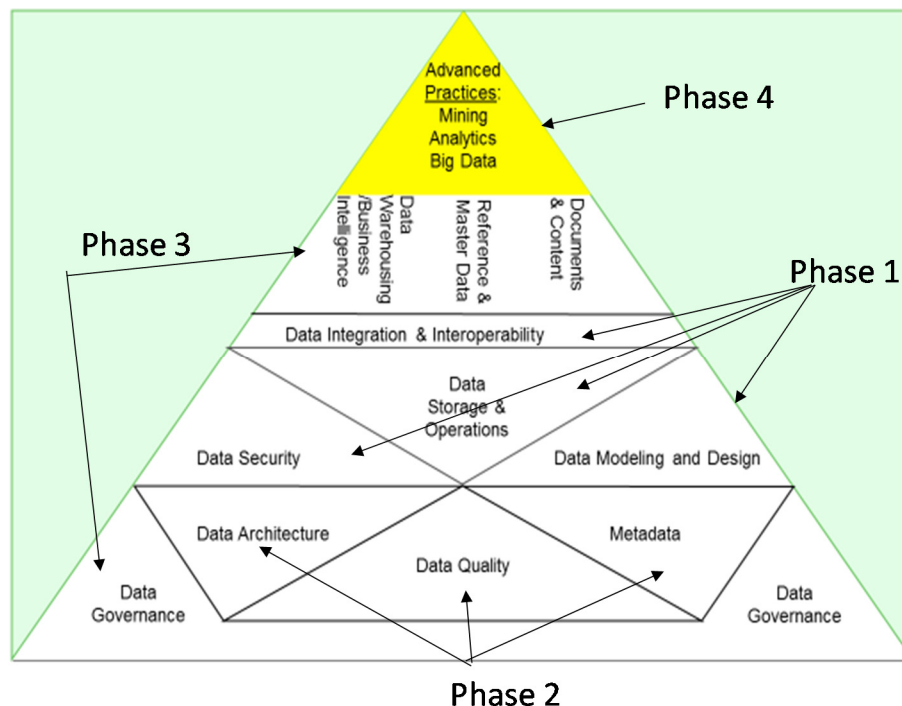


Figure 8 Purchased or Built Database Capability¹⁷

Aiken's pyramid draws from the DAMA Wheel, but also informs it by showing the relation between the Knowledge Areas. They are not all interchangeable; they have various kinds of interdependencies. The Pyramid framework has two drivers. First, the idea of building on a foundation, using components that need to be in the right places to support each other. Second, the somewhat contradictory idea that these may be put in place in an arbitrary order.

3.5 DAMA Data Management Framework Evolved

Aiken's pyramid describes how organizations evolve toward better data management practices. Another way to look at the DAMA Knowledge Areas is to explore the dependencies between them. Developed by Sue Geuens, the framework in Figure 9 recognizes that Business Intelligence and Analytic functions have dependencies on all other data management functions. They depend directly on Master Data and data warehouse solutions. But those, in turn, are dependent on feeding systems and applications. Reliable Data Quality, data design, and data interoperability practices are at the foundation of reliable systems and applications. In addition, data governance, which within this

¹⁷ Golden Pyramid figure copyright Data Blueprint, used with permission.

model includes Metadata Management, data security, Data Architecture and Reference Data Management, provides a foundation on which all other functions are dependent.

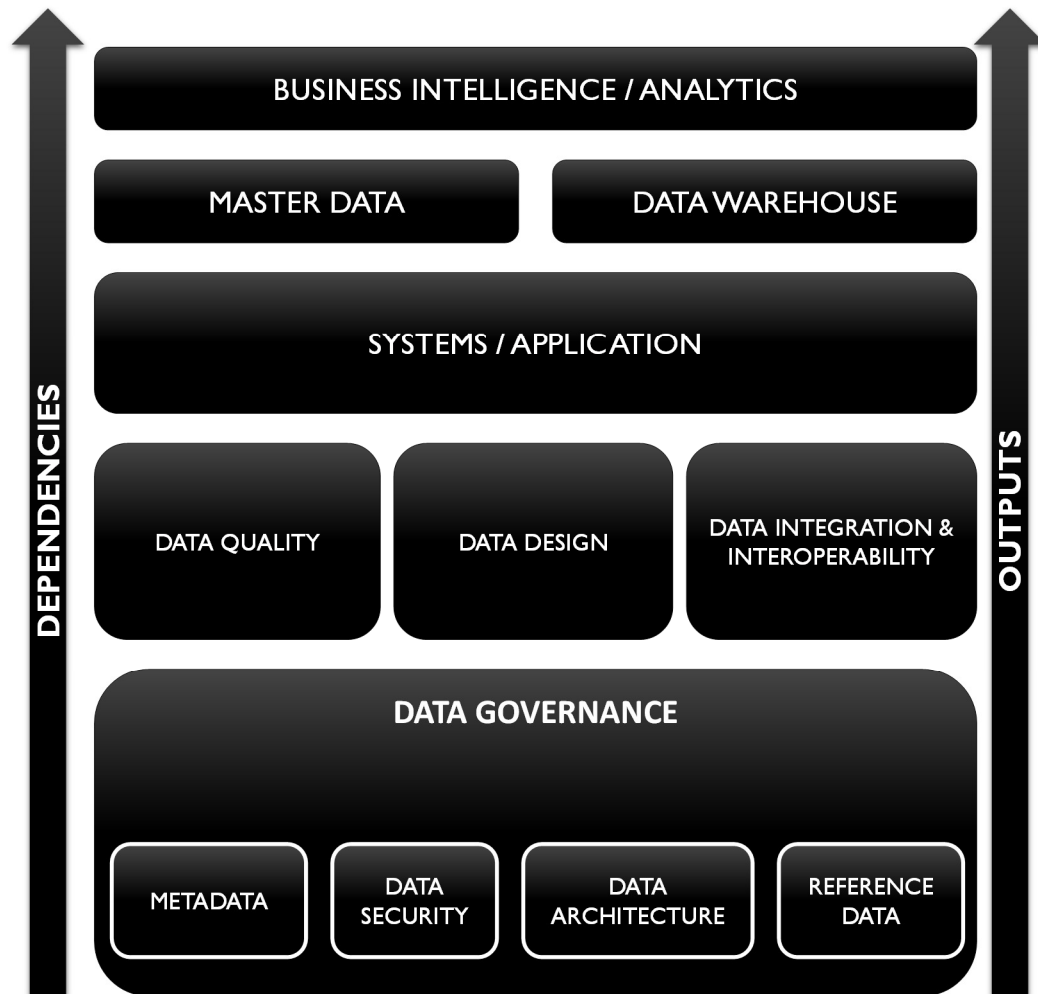


Figure 9 DAMA Functional Area Dependencies

A third alternative to DAMA Wheel is depicted in Figure 10. This also draws on architectural concepts to propose a set of relationships between the DAMA Knowledge Areas. It provides additional detail about the content of some Knowledge Areas in order to clarify these relationships.

The framework starts with the guiding purpose of data management: To enable organizations to get value from their data assets as they do from other assets. Deriving value requires lifecycle management, so data management functions related to the data lifecycle are depicted in the center of the diagram. These include planning and designing for reliable, high quality data; establishing processes and functions through which data can be enabled for use and also maintained; and, finally, using the data in various types of analysis and through those processes, enhancing its value.

The lifecycle management section depicts the data management design and operational functions (modeling, architecture, storage and operations, etc.) that are required to support traditional uses of data (Business Intelligence, document and content management). It also recognizes emerging data management functions (Big Data storage)

that support emerging uses of data (Data Science, predictive analytics, etc.). In cases where data is truly managed as an asset, organizations may be able to get direct value from their data by selling it to other organizations (data monetization).

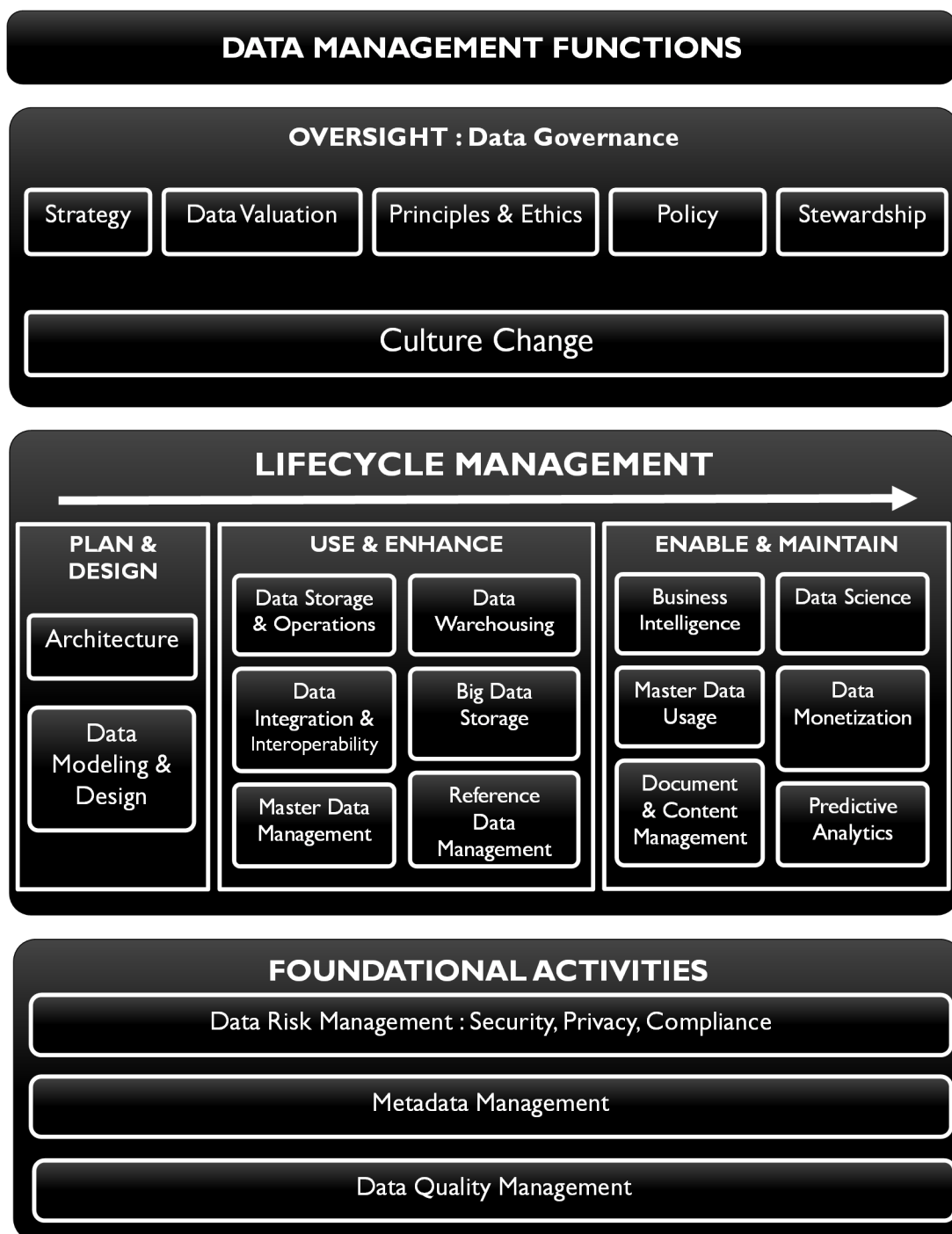


Figure 10 DAMA Data Management Function Framework

Organizations that focus only on direct lifecycle functions will not get as much value from their data as those that support the data lifecycle through foundational and oversight activities. Foundational activities, like data risk

management, Metadata, and Data Quality management, span the data lifecycle. They enable better design decisions and make data easier to use. If these are executed well, data is less expensive to maintain, data consumers have more confidence in it, and the opportunities for using it expand.

To successfully support data production and use and to ensure that foundational activities are executed with discipline, many organizations establish oversight in the form of data governance. A data governance program enables an organization to be data-driven, by putting in place the strategy and supporting principles, policies, and stewardship practices that ensure the organization recognizes and acts on opportunities to get value from its data. A data governance program should also engage in organizational change management activities to educate the organization and encourage behaviors that enable strategic uses of data. Thus, the necessity of culture change spans the breadth of data governance responsibilities, especially as an organization matures its data management practices.

The DAMA Data Management Framework can also be depicted as an evolution of the DAMA Wheel, with core activities surrounded by lifecycle and usage activities, contained within the strictures of governance. (See Figure 11.)

Core activities, including Metadata Management, Data Quality Management, and data structure definition (architecture) are at the center of the framework.

Lifecycle management activities may be defined from a planning perspective (risk management, modeling, data design, Reference Data Management) and an enablement perspective (Master Data Management, data technology development, data integration and interoperability, data warehousing, and data storage and operations).

Usages emerge from the lifecycle management activities: Master data usage, Document and content management, Business Intelligence, Data Science, predictive analytics, data visualization. Many of these create more data by enhancing or developing insights about existing data. Opportunities for data monetization may be identified as uses of data.

Data governance activities provide oversight and containment, through strategy, principles, policy, and stewardship. They enable consistency through data classification and data valuation.

The intention in presenting different visual depictions of the DAMA Data Management Framework is to provide additional perspective and to open discussion about how to apply the concepts presented in DMBOK. As the importance of data management grows, such frameworks become useful communications tools both within the data management community and between the data management community and our stakeholders.

4. DAMA and the DMBOK

While data management presents many challenges, few of them are new. Since at least the 1980s, organizations have recognized that managing data is central to their success. As our ability and desire to create and exploit data has increased, so too has the need for reliable data management practices.

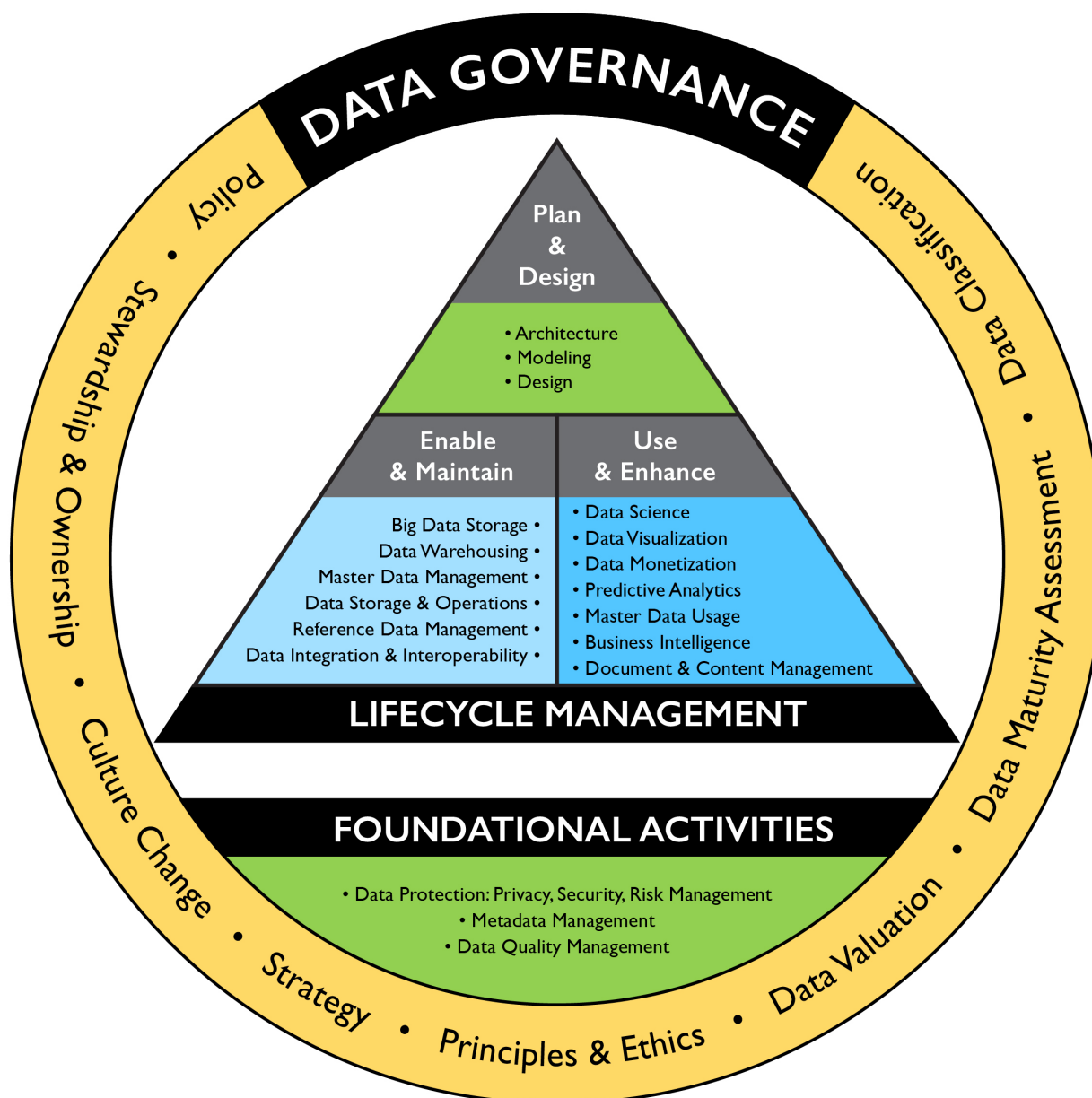


Figure 11 DAMA Wheel Evolved

DAMA was founded to address these challenges. The DMBOK, an accessible, authoritative reference book for data management professionals, supports DAMA's mission by:

- **Providing a functional framework** for the implementation of enterprise data management practices; including guiding principles, widely adopted practices, methods and techniques, functions, roles, deliverables and metrics.
- **Establishing a common vocabulary** for data management concepts and serving as the basis for best practices for data management professionals.

- **Serving as the fundamental reference guide** for the CDMP (Certified Data Management Professional) and other certification exams.

The DMBOK is structured around the eleven Knowledge Areas of the DAMA-DMBOK Data Management Framework (also known as the DAMA Wheel, see Figure 5). Chapters 3 – 13 are focused on Knowledge Areas. Each Knowledge Area chapter follows a common structure:

1. Introduction
 - Business Drivers
 - Goals and Principles
 - Essential Concepts
2. Activities
3. Tools
4. Techniques
5. Implementation Guidelines
6. Relation to Data Governance
7. Metrics

Knowledge Areas describe the scope and context of sets of data management activities. Embedded in the Knowledge Areas are the fundamental goals and principles of data management. Because data moves horizontally within organizations, Knowledge Area activities intersect with each other and with other organizational functions.

1. **Data Governance** provides direction and oversight for data management by establishing a system of decision rights over data that accounts for the needs of the enterprise. (Chapter 3)
2. **Data Architecture** defines the blueprint for managing data assets by aligning with organizational strategy to establish strategic data requirements and designs to meet these requirements. (Chapter 4)
3. **Data Modeling and Design** is the process of discovering, analyzing, representing, and communicating data requirements in a precise form called the *data model*. (Chapter 5)
4. **Data Storage and Operations** includes the design, implementation, and support of stored data to maximize its value. Operations provide support throughout the data lifecycle from planning for to disposal of data. (Chapter 6)
5. **Data Security** ensures that data privacy and confidentiality are maintained, that data is not breached, and that data is accessed appropriately. (Chapter 7)
6. **Data Integration and Interoperability** includes processes related to the movement and consolidation of data within and between data stores, applications, and organizations. (Chapter 8)
7. **Document and Content Management** includes planning, implementation, and control activities used to manage the lifecycle of data and information found in a range of unstructured media, especially documents needed to support legal and regulatory compliance requirements. (Chapter 9)

8. **Reference and Master Data** includes ongoing reconciliation and maintenance of core critical shared data to enable consistent use across systems of the most accurate, timely, and relevant version of truth about essential business entities. (Chapter 10)
9. **Data Warehousing and Business Intelligence** includes the planning, implementation, and control processes to manage decision support data and to enable knowledge workers to get value from data via analysis and reporting. (Chapter 11)
10. **Metadata** includes planning, implementation, and control activities to enable access to high quality, integrated Metadata, including definitions, models, data flows, and other information critical to understanding data and the systems through which it is created, maintained, and accessed. (Chapter 12)
11. **Data Quality** includes the planning and implementation of quality management techniques to measure, assess, and improve the fitness of data for use within an organization. (Chapter 13)

In addition to chapters on the Knowledge Areas, the DAMA-DMBOK contains chapters on the following topics:

- **Data Handling Ethics** describes the central role that data ethics plays in making informed, socially responsible decisions about data and its uses. Awareness of the ethics of data collection, analysis, and use should guide all data management professionals. (Chapter 2)
- **Big Data and Data Science** describes the technologies and business processes that emerge as our ability to collect and analyze large and diverse data sets increases. (Chapter 14.)
- **Data Management Maturity Assessment** outlines an approach to evaluating and improving an organization's data management capabilities. (Chapter 15)
- **Data Management Organization and Role Expectations** provide best practices and considerations for organizing data management teams and enabling successful data management practices. (Chapter 16)
- **Data Management and Organizational Change Management** describes how to plan for and successfully move through the cultural changes that are necessary to embed effective data management practices within an organization. (Chapter 17)

How a particular organization manages its data depends on its goals, size, resources, and complexity, as well as its perception of how data supports its overall strategy. Most enterprises do not perform all the activities described in each Knowledge Area. However, understanding the wider context of data management will enable organizations to make better decisions about where to focus as they work to improve practices within and across these related functions.

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Data Handling Ethics

1. Introduction

Defined simply, *ethics* are principles of behavior based on ideas of right and wrong. Ethical principles often focus on ideas such as fairness, respect, responsibility, integrity, quality, reliability, transparency, and trust. Data handling ethics are concerned with how to procure, store, manage, use, and dispose of data in ways that are aligned with ethical principles. Handling data in an ethical manner is necessary to the long-term success of any organization that wants to get value from its data. Unethical data handling can result in the loss of reputation and customers, because it puts at risk people whose data is exposed. In some cases, unethical practices are also illegal.¹⁸ Ultimately, for data management professionals and the organizations for which they work, data ethics are a matter of social responsibility.

The ethics of data handling are complex, but they center on several core concepts:

- **Impact on people:** Because data represents characteristics of individuals and is used to make decisions that affect people's lives, there is an imperative to manage its quality and reliability.
- **Potential for misuse:** Misusing data can negatively affect people and organizations, so there is an ethical imperative to prevent the misuse of data.
- **Economic value of data:** Data has economic value. Ethics of data ownership should determine how that value can be accessed and by whom.

Organizations protect data based largely on laws and regulatory requirements. Nevertheless, because data represents people (customers, employees, patients, vendors, etc.), data management professionals should recognize that there are ethical (as well as legal) reasons to protect data and ensure it is not misused. Even data that does not directly represent individuals can still be used to make decisions that affect people's lives.

There is an ethical imperative not only to protect data, but also to manage its quality. People making decisions, as well as those impacted by decisions, expect data to be complete and accurate. From both a business and a technical perspective, data management professionals have an ethical responsibility to manage data in a way that reduces the

¹⁸ HIPAA (Health Insurance Portability and Accountability Act) in the US, PIPEDA (Personal Information Protection and Electronic Documents Act) in Canada, the EU General Data Protection Regulation (GDPR) and other data protection / information privacy laws describe obligations toward the handling of personal identifying data (e.g., name, addresses, religious affiliation, or sexual orientation) and privacy (access or restriction to this information).

risk that it may misrepresent, be misused, or be misunderstood. This responsibility extends across the data lifecycle, from creation to destruction of data.

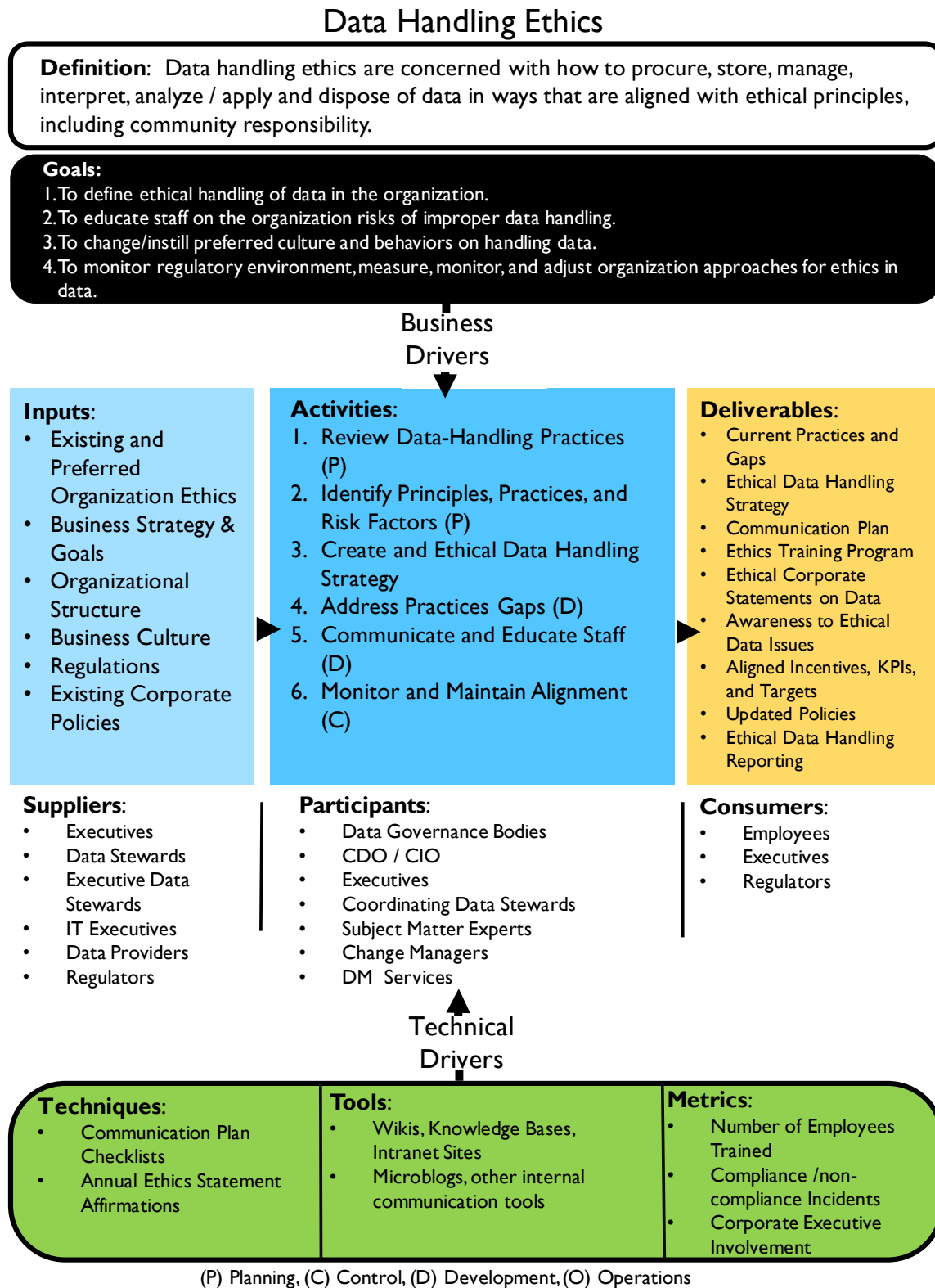


Figure 12 Context Diagram: Data Handling Ethics

Unfortunately, many organizations fail to recognize and respond to the ethical obligations inherent in data management. They may adopt a traditional technical perspective and profess not to understand the data; or they assume that if they follow the letter of the law, they have no risk related to data handling. This is a dangerous assumption.

The data environment is evolving rapidly. Organizations are using data in ways they would not have imagined even a few years ago. While laws codify some ethical principles, legislation cannot keep up with the risks associated with evolution of the data environment. Organizations must recognize and respond to their ethical obligation to protect data entrusted to them by fostering and sustaining a culture that values the ethical handling of information.

2. Business Drivers

Like W. Edward Deming's statements on quality, ethics means "doing it right when no one is looking." An ethical approach to data use is increasingly being recognized as a competitive business advantage (Hasselbalch and Tranberg, 2016). Ethical data handling can increase the trustworthiness of an organization and the organization's data and process outcomes. This can create better relationships between the organization and its stakeholders. Creating an ethical culture entails introducing proper governance, including institution of controls to ensure that both intended and resulting outcomes of data processing are ethical and do not violate trust or infringe on human dignity.

Data handling doesn't happen in a vacuum, and customers and stakeholders expect ethical behavior and outcomes from businesses and their data processes. Reducing the risk that data for which the organization is responsible will be misused by employees, customers, or partners is a primary reason for an organization to cultivate ethical principles for data handling. There is also an ethical responsibility to secure data from criminals (i.e., to protect against hacking and potential data breaches. (See Chapter 7.)

Different models of data ownership influence the ethics of data handling. For example, technology has improved the ability of organizations to share data with each other. This ability means organizations need to make ethical decisions about their responsibility for sharing data that does not belong to them.

The emerging roles of Chief Data Officer, Chief Risk Officer, Chief Privacy Officer, and Chief Analytics Officer are focused on controlling risk by establishing acceptable practices for data handling. But responsibility extends beyond people in these roles. Handling data ethically requires organization-wide recognition of the risks associated with misuse of data and organizational commitment to handling data based on principles that protect individuals and respect the imperatives related to data ownership.

3. Essential Concepts

3.1 Ethical Principles for Data

The accepted tenets of bioethics, which focus on preserving human dignity, provide a good general starting point for principles of data ethics. For example, the Belmont Principles for medical research may be adapted in Information Management disciplines (US-HSS, 1979).

- **Respect for Persons:** This principle reflects the fundamental ethical requirement that people be treated in a way that respects their dignity and autonomy as human individuals. It also requires that in cases where people have ‘diminished autonomy’, extra care be taken to protect their dignity and rights.

When we consider data as an asset, do we keep in mind that data also affects, represents, or touches people? Personal data is different from other raw ‘assets’, like oil or coal. Unethical use of personal data can directly influence people’s interactions, employment opportunities, and place in the community. Do we design information systems in a way that limits autonomy or freedom of choice? Have we considered how processing data may affect people with mental or physical disabilities? Have we accounted for how they will access and utilize data? Does data processing take place on the basis of informed, valid consent?

- **Beneficence:** This principle has two elements: first, do not harm; second, maximize possible benefits and minimize possible harms.

The ethical principle of ‘do not harm’ has a long history in medical ethics, but also has clear application in the context of data and information management. Ethical data and information practitioners should identify stakeholders and consider the outcomes of data processing and work to maximize benefit and minimize risk of harm caused by the processes designed. Is a process designed in a way that assumes a zero-sum outcome rather than a win-win situation? Is data processing unnecessarily invasive and is there a less risky way to meet the requirements of the business need? Is the data handling in question lacking transparency in a way that might hide possible harm to people?

- **Justice:** This principle considers the fair and equitable treatment of people.

Some questions that might be asked regarding this principle: Are people or groups of people being treated unequally under similar circumstances? Does the outcome of a process or algorithm result in effects that disproportionately benefit or harm a certain group of people? Is machine learning being trained using datasets that contain data inadvertently reinforcing cultural prejudices?

The United States Department of Homeland Security’s Menlo Report adapts the Belmont Principles to Information and Communication Technology Research, adding a fourth principle: Respect for Law and Public Interest (US-DHS, 2012).

In 2015, the European Data Protection Supervisor published an opinion on digital ethics highlighting the “engineering, philosophical, legal, and moral implications” of developments in data processing and Big Data. It

called for a focus on data processing that upholds human dignity, and set out four pillars required for an information ecosystem that ensures ethical treatment of data (EDPS, 2015):

- Future-oriented regulation of data processing and respect for the rights to privacy and to data protection
- Accountable controllers who determine personal information processing
- Privacy conscious engineering and design of data processing products and services
- Empowered individuals

These principles map broadly to the principle set out in the Belmont Report, focusing on promoting human dignity and autonomy. The EDPS states that privacy is a fundamental human right. It challenges innovators to see dignity, privacy, and autonomy as a platform on which a sustainable digital environment is shaped, rather than an obstacle to development, and calls for transparency and communication with stakeholders.

Data Governance is a vital tool for ensuring these principles are considered in deciding who can do what with which data and under what circumstances processing is appropriate or necessary. The ethical impacts and risks of data processing on all stakeholders must be considered by practitioners, and managed in a similar manner to data quality.

3.2 Principles Behind Data Privacy Law

Public policy and law try to codify right and wrong based on ethical principles. But they cannot codify every circumstance. For example, privacy laws in the European Union, Canada, and the United States show different approaches to codifying data ethics. These principles can also provide a framework for organizational policy.

Privacy law is not new. Privacy and information privacy as concepts are firmly linked to the ethical imperative to respect human rights. In 1890, American legal scholars Samuel Warren and Louis Brandeis described privacy and information privacy as human rights with protections in common law that underpin several rights in the US constitution. In 1973, a code of Fair Information Practice was proposed, and the concept of information privacy as a fundamental right was reaffirmed in the US Privacy Act of 1974, which states that “the right to privacy is a personal and fundamental right protected by the Constitution of the United States”.

In the wake of human rights violations during the Second World War, the European Convention of Human Rights (1950) established both the general right to privacy and the specific right to information privacy (or the right to protection of one’s personal data) as human rights which are fundamental to upholding the right to Human Dignity. In 1980, the Organization for Economic Co-operation and Development (OECD) established Guidelines and Principles for Fair Information Processing that became the basis for the European Union’s data protection laws.

OECD’s eight core principles, the Fair Information Processing Standards, are intended to ensure that personal data is processed in a manner that respects individuals’ right to privacy. They include: limitations on data collection; an expectation that data will be of high quality; the requirement that when data is collected, it is done for a specific purpose; limitations on data usage; security safeguards; an expectation of openness and transparency; the right of an individual to challenge the accuracy of data related to himself or herself; and accountability for organizations to follow the guidelines.

The OECD principles have since been superseded by principles underlying the General Data Protection Regulation of the EU, (GDPR, 2016). See Table 1.

Table 1 GDPR Principles

GDPR Principle	Description of Principle
Fairness, Lawfulness, Transparency	Personal data shall be processed lawfully, fairly, and in a transparent manner in relation to the data subject.
Purpose Limitation	Personal data must be collected for specified, explicit, and legitimate purposes, and not processed in a manner that is incompatible with those purposes.
Data Minimization	Personal data must be adequate, relevant, and limited to what is necessary in relation to the purposes for which they are processed.
Accuracy	Personal data must be accurate, and where necessary, kept up-to-date. Every reasonable step must be taken to ensure that personal data that are inaccurate, having regard to the purpose for which they are processed, are erased or rectified without delay.
Storage Limitation	Data must be kept in a form that permits identification of data subjects [individuals] for no longer than is necessary for the purposes for which the personal data are processed.
Integrity and Confidentiality	Data must be processed in a manner that ensures appropriate security of the personal data, including protection against unauthorized or unlawful processing and against accidental loss, destruction or damage, using appropriate technical or organizational measures.
Accountability	Data Controllers shall be responsible for, and be able to demonstrate compliance with [these principles].

These principles are balanced by and support certain qualified rights individuals have to their data, including the rights to access, rectification of inaccurate data, portability, the right to object to processing of personal data that may cause damage or distress, and erasure. When processing of personal data is done based on consent, that consent must be an affirmative action that is freely given, specific, informed, and unambiguous. The GDPR requires effective governance and documentation to enable and demonstrate compliance and mandates Privacy by Design.

Canadian privacy law combines a comprehensive regime of privacy protection with industry self-regulation. PIPEDA (Personal Information Protection and Electronic Documents Act) applies to every organization that collects, uses, and disseminates personal information in the course of commercial activities. It stipulates rules, with exceptions, that organizations must follow in their use of consumers' personal information. Table 2 describes statutory obligations based on PIPEDA.¹⁹

In Canada, the Federal Privacy Commissioner has the sole responsibility for handling privacy complaints against organizations. However, they fill an ombudsman role; their decisions are only recommendations (not legally binding and with no precedential value, even within the commissioner's office).

¹⁹ <http://bit.ly/2tNM53c>.

Table 2 Canadian Privacy Statutory Obligations

PIPEDA Principle	Description of Principle
Accountability	An organization is responsible for personal information under its control and must designate an individual to be accountable for the organization's compliance with the principle.
Identifying Purposes	An organization must identify the purposes for which personal information is collected at or before the time the information is collected.
Consent	An organization must obtain the knowledge and consent of the individual for the collection, use, or disclosure of personal information, except where inappropriate.
Limiting Collection, Use, Disclosure, and Retention	The collection of personal information must be limited to that which is necessary for the purposes identified by the organization. Information shall be collected by fair and lawful means. Personal information shall not be used or disclosed for purposes other than those for which it was collected, except with the consent of the individual or as required by law. Personal information shall be retained only as long as necessary for the fulfillment of those purposes.
Accuracy	Personal information must be as accurate, complete, and up-to-date as is necessary for the purposes for which it is to be used.
Safeguards	Personal information must be protected by security safeguards appropriate to the sensitivity of the information.
Openness	An organization must make specific information about its policies and practices relating to the management of their personal information readily available to individuals.
Individual Access	Upon request, an individual shall be informed of the existence, use, and disclosure of his or her personal information, and shall be given access to that information. An individual shall be able to challenge the accuracy and completeness of the information and have it amended as appropriate.
Compliance Challenges	An individual shall be able to address a challenge concerning compliance with the above principles to the designated individual or individuals accountable for the organization's compliance.

In March 2012, the US Federal Trade Commission (FTC) issued a report recommending organizations design and implement their own privacy programs based on best practices described in the report (i.e., Privacy by Design) (FTC 2012). The report reaffirms the FTC's focus on Fair Information Processing Principles (see Table 3).

Table 3 United States Privacy Program Criteria

Principle	Description of Principle
Notice / Awareness	Data collectors must disclose their information practices before collecting personal information from consumers.
Choice / Consent	Consumers must be given options with respect to whether and how personal information collected from them may be used for purposes beyond those for which the information was provided.
Access / Participation	Consumers should be able to view and contest the accuracy and completeness of data collected about them.
Integrity / Security	Data collectors must take reasonable steps to assure that information collected from consumers is accurate and secure from unauthorized use.
Enforcement / Redress	The use of a reliable mechanism to impose sanctions for noncompliance with these fair information practices.

These principles are developed to embody the concepts in the OECD Fair Information Processing Guidelines, including emphasis on data minimization (reasonable collection limitation) and storage limitation (sound retention), accuracy, and the requirement that companies must provide reasonable security for consumer data. Other focuses for fair information practices include:

- Simplified consumer choice to reduce the burden placed on consumers
- The recommendation to maintain comprehensive data management procedure throughout the information lifecycle
- Do Not Track option
- Requirements for affirmative express consent
- Concerns regarding the data collection capabilities of large platform providers; transparency and clear privacy notices and policies
- Individuals' access to data
- Educating consumers about data privacy practices
- Privacy by Design

There is a global trend towards increasing legislative protection of individuals' information privacy, following the standards set by EU legislation. Laws around the world place different kinds of restrictions on the movement of data across international boundaries. Even within a multinational organization, there will be legal limits to sharing information globally. It is therefore important that organizations have policies and guidelines that enable staff to follow legal requirements as well as use data within the risk appetite of the organization.

3.3 Online Data in an Ethical Context

There are now emerging dozens of initiatives and programs designed to create a codified set of principles to inform ethical behaviors online in the United States (Davis, 2012). Topics include:

- **Ownership of data:** The rights to control one's personal data in relation to social media sites and data brokers. Downstream aggregators of personal data can embed data into deep profiles that individuals are not aware of.
- **The Right to be Forgotten:** To have information about an individual be erased from the web, particularly to adjust online reputation. This topic is part of data retention practices in general.
- **Identity:** Having the right to expect one identity and a correct identity, and to opt for a private identity.
- **Freedom of speech online:** Expressing one's opinions versus bullying, terror inciting, 'trolling,' or insulting.

3.4 Risks of Unethical Data Handling Practices

Most people who work with data know that it is possible to use data to misrepresent facts. The classic book *How to Lie with Statistics* by Darrell Huff (1954) describes a range of ways that data can be used to misrepresent facts

while creating a veneer of factuality. Methods include judicious data selection, manipulation of scale, and omission of some data points. These approaches are still at work today.

One way to understand the implications of ethical handling of data is to examine practices that most people would agree are unethical. Ethical data handling entails a positive duty to handle data according to ethical principles such as trustworthiness. Ensuring data is trustworthy may include measuring is against Data Quality dimensions such as accuracy and timeliness. It also includes a base level of truthfulness and transparency – not using data to lie or mislead, and being transparent regarding the sources, uses, and intent behind an organization’s data handling. The following scenarios describe unethical data practices that violate these principles among others.

3.4.1 Timing

It is possible to lie through omission or inclusion of certain data points in a report or activity based on timing. Equity market manipulation through ‘end of day’ stock trades can artificially raise a stock price at closing of the market giving an artificial view of the stock’s worth. This is called market timing and is illegal.

Business Intelligence staff may be the first to notice anomalies. In fact, they are now seen as valuable players in the stock trading centers of the world recreating trading patterns looking for such problems as well as analyzing reports and reviewing and monitoring rules and alerts. Ethical Business Intelligence staff may need to alert appropriate governance or management functions to such anomalies.

3.4.2 Misleading Visualizations

Charts and graphs can be used to present data in a misleading manner. For instance, changing scale can make a trend line look better or worse. Leaving data points out, comparing two facts without clarifying their relationship, or ignoring accepted visual conventions (such as that the numbers in a pie chart representing percentages must add up to 100 and only 100), can also be used to trick people into interpreting visualizations in ways that are not supported by the data itself.²⁰

3.4.3 Unclear Definitions or Invalid Comparisons

A US news outlet reported, based on 2011 US Census Bureau data, that 108.6 million people in the US were on welfare yet only 101.7 million people had full time jobs, making it seem that a disproportionate percentage of the overall population was on welfare.²¹ Media Matters explained the discrepancy: The 108.6 million figure for the number of “people on welfare” comes from a Census Bureau’s account ... of participation in means-tested

²⁰ How To Statistics (Website). *Misleading Graphs: Real Life Examples*. 24 January 2014. <http://bit.ly/1jRLgRH> See also io9 (Website). *The Most Useless and Misleading Infographics on the Internet*. <http://bit.ly/1YDgURI> See <http://bit.ly/2tNktve> Google “misleading data visualization” for additional examples. For counter examples, i.e., visuals with an ethical base, see Tufte (2001).

²¹ As of 2015, the overall population of the US is estimated to be 321.4 million people. <http://bit.ly/2iMIP58>

programs, which include “anyone residing in a household in which one or more people received benefits” in the fourth quarter of 2011, thus including individuals who did not themselves receive government benefits. On the other hand, the “people with a full time job” figure ... included only individuals who worked, not individuals residing in a household where at least one person works.²²

The ethical thing to do, in presenting information, is to provide context that informs its meaning, such as a clear, unambiguous definition of the population being measured and what it means to be “on welfare.” When required context is left out, the surface of the presentation may imply meaning that the data does not support. Whether this effect is gained through the intent to deceive or through simply clumsiness, it is an unethical use of data.

It is also simply necessary, from an ethical perspective, not to misuse statistics.

Statistical ‘smoothing’ of numbers over a period could completely change perception of the number. ‘Data mining snooping’ is a recently coined term for a phenomenon in data mining statistical investigations where exhaustive correlations are performed on a data set, essentially over training a statistical model. Because of the behavior of ‘statistical significance’, it is reasonable to expect some statistically significant-looking results that are actually random results. The untrained can be misled. This is common in the financial and medical sectors (Jensen, 2000; ma.utexas.edu, 2012).²³

3.4.4 Bias

Bias refers to an inclination of outlook. On the personal level, the term is associated with unreasoned judgments or prejudices. In statistics, bias refers to deviations from expected values. These are often introduced through systematic errors in sampling or data selection.²⁴ Bias can be introduced at different points in the data lifecycle: when data is collected or created, when it is selected for inclusion in analysis, through the methods by which it is analyzed, and in how the results of analysis are presented.

The ethical principle of justice creates a positive duty to be aware of possible biases that might influence data collection, processing, analysis, or interpretation. This is particularly important in the case of large-scale data processing that might disproportionately affect groups of people that have been historically subjected to prejudice or unfair treatment. Using data without addressing the ways in which bias may be introduced can compound prejudice while reducing transparency in process, giving the resulting outcomes the veneer of impartiality or neutrality when they are not neutral. There are several types of bias:

- **Data Collection for pre-defined result:** The analyst is pressured to collect data and produce results in order to reach a pre-defined conclusion, rather than as an effort to draw an objective conclusion.

²² <http://mm4a.org/2spKToU> The example also demonstrates misleading visuals, as on the bar graph, the 108.6 million bar was shown as approximately 5 times larger than the 101.7 million column.

²³ See also numerous articles by W. Edwards Deming at: <http://bit.ly/2tNnIZh>

²⁴ <http://bit.ly/2lOzJqU>

- **Biased use of data collected:** Data may be collected with limited bias, but an analyst is pressured to use it to confirm a pre-determined approach. Data may even be manipulated to this end (i.e., some data may be discarded if it does not confirm the approach).
- **Hunch and search:** The analyst has a hunch and wants to satisfy that hunch, but uses only the data that confirms the hunch and does not account for other possibilities that the data may surface.
- **Biased sampling methodology:** Sampling is often a necessary part of data collection. But bias can be introduced by the method used to select the sample set. It is virtually impossible for humans to sample without bias of some sort. To limit bias, use statistical tools to select samples and establish adequate sample sizes. Awareness of bias in data sets used for training is particularly important.
- **Context and Culture:** Biases are often culturally or contextually-based, so stepping outside that culture or context is required for a neutral look at the situation.

Questions of bias are dependent on many factors, such as the type of data processing in question, the stakeholders involved, how data sets are populated, the business need being fulfilled, and the expected outcomes of the process. However, it is not always possible or even desirable to remove all bias. Business bias against poor customers (customers with whom no further business is sought) is a foundational piece to many scenarios built by business analysts; they are de-selected from samples, or ignored in the analysis. In such a case, analysts should document the criteria they used to define the population they are studying. In contrast, predictive algorithms determining ‘criminal risk’ of individuals or predictive policing sending resources to specific neighborhoods would have a much higher risk of violating ethical principles of justice or fairness, and should have greater precautions to ensure algorithmic transparency and accountability and to counter bias in data sets training any predictive algorithms.²⁵

3.4.5 Transforming and Integrating Data

Data integration presents ethical challenges because data is changed as it moves from system to system. If data is not integrated with care, it presents risk for unethical or even illegal data handling. These ethical risks intersect with fundamental problems in data management, including:

- **Limited knowledge of data’s origin and lineage:** If an organization does not know where data came from and how it has changed as it has moved between systems, then the organization cannot prove that the data represents what they claim it represents.
- **Data of poor quality:** Organizations should have clear, measurable standards for data quality, and should measure their data to confirm that it meets standards for quality. Without this confirmation, an organization cannot vouch for the data and data consumers may be at risk or put others at risk when they use the data.

²⁵ For examples of machine learning bias see Brennan (2015) and the Ford Foundation and ProPublica websites. In addition to bias, there is the problem of opaqueness. As predictive algorithms of learning machines become more complex, it is difficult to track the logic and lineage of their decisions. See Lewis and Monett (2017). <http://bit.ly/1Om41ap>; <http://bit.ly/2oYmNRu>.

- **Unreliable Metadata:** Data consumers depend on reliable Metadata, including consistent definitions of individual data elements, documentation of data's origin, and documentation of lineage (e.g., rules by which data is integrated). Without reliable Metadata, data may be misunderstood and potentially misused. In cases where data may move between organizations and especially where it may move across borders, Metadata should include tags that indicate its provenance, who owns it, and if it requires specific protection.
- **No documentation of data remediation history:** Organizations should also have auditable information related to the ways data has been changed. Even if the intention of data remediation is to improve the quality of data, doing so may be illegal. Data remediation should always follow a formal, auditable change control process.

3.4.6 Obfuscation / Redaction of Data

Obfuscating or redacting data is the practice of making information anonymous, or removing sensitive information. But obfuscation alone may not be sufficient to protect data if a downstream activity (analysis or combination with other datasets) can expose the data. This risk is present in the following instances:

- **Data aggregation:** When aggregating data across some set of dimensions, and removing identifying data, a dataset can still serve an analytic purpose without concern for disclosing personal identifying information (PII). Aggregations into geographic areas are a common practice (see Chapters 7 and 14).
- **Data marking:** Data marking is used to classify data sensitivity (secret, confidential, personal, etc.) and to control release to appropriate communities such as the public or vendors, or even vendors from certain countries or other community considerations.
- **Data masking:** Data masking is a practice where only appropriate submitted data will unlock processes. Operators cannot see what the appropriate data might be; they simply type in responses given to them, and if those responses are correct, further activities are permitted. Business processes using data masking include outsourced call centers, or sub-contractors who should only have partial access to information.

The use of extremely large data sets in Data Science analyses raises practical rather than merely theoretical concerns about the effectiveness of anonymization. Within large data sets, it is possible to combine data in ways enable individuals to be specifically identified, even if input data sets have been anonymized. The first concern when data lands in a data lake is to analyze it for sensitive data and apply accepted protection methods. These alone may not offer enough safeguard, however; this is why it is vital that organizations have strong governance and a commitment to ethical data handling. (See Chapter 14.)

3.5 Establishing an Ethical Data Culture

Establishing a culture of ethical data handling requires understanding existing practices, defining expected behaviors, codifying these in policies and a code of ethics, and providing training and oversight to enforce expected

behaviors. As with other initiatives related to governing data and to changing culture, this process requires strong leadership.

Ethical handling of data obviously includes following the law, but also influences how data is analyzed and interpreted, as well as how it is leveraged internally and externally. An organizational culture that clearly values ethical behavior will not only have codes of conduct, but will ensure that clear communication and governance controls are in place to support employees with queries and proper escalation paths so that if employees become aware of unethical practices or ethical risk they are able to highlight the problem or stop the process without fear of retaliation. Improving an organization's ethical behavior regarding data requires a formal Organizational Change Management (OCM) process. (See Chapter 17.)

3.5.1 Review Current State Data Handling Practices

The first step to improvement is understanding the current state. The purpose of reviewing existing data handling practices is to understand the degree to which they are directly and explicitly connected to ethical and compliance drivers. This review should also identify how well employees understand the ethical implications of existing practices in building and preserving the trust of customers, partners, and other stakeholders. The deliverable from the review should document ethical principles that underlie the organization's collection, use, and oversight of data, throughout the data lifecycle, including data sharing activities.

3.5.2 Identify Principles, Practices, and Risk Factors

The purpose of formalizing ethical practices around data handling is to reduce the risk that data might be misused and cause harm to customers, employees, vendors, other stakeholders, or the organization as a whole. An organization trying to improve its practices should be aware of general principles, such as the necessity of protecting the privacy of individuals, as well as industry-specific concerns, such as the need to protect financial or health-related information.

An organization's approach to data ethics must align with legal and regulatory compliance requirements. For example, organizations that operate globally need to have a broad knowledge of the ethical principles at the foundation of the laws of the countries in which they operate, as well as specific knowledge of the agreements between countries. In addition, most organizations have specific risks, which may be related to their technology footprint, their rate of employee turnover, the means by which they collect customer data, or other factors.

Principles should be aligned with risks (bad things that can happen if the principles are not adhered to) and practices (the right ways of doing things so that risks are avoided). Practices should be supported by controls, as illustrated in the following example:

- **Guiding principle:** People have a right to privacy with respect to information about their health. Therefore, the personal health data of patients should not be accessed except by people who are authorized to access it as part of caring for patients.

- **Risk:** If there is wide access to the personal health data of patients, then information about individuals could become public knowledge, thereby jeopardizing their right to privacy.
- **Practice:** Only nurses and doctors will be allowed to access the personal health data of patients and only for purposes of providing care.
- **Control:** There will be an annual review of all users of the systems that contain personal health information of patients to ensure that only those people who need to have access do have access.

3.5.3 Create an Ethical Data Handling Strategy and Roadmap

After a review of current state and the development of a set of principles, an organization can formalize a strategy to improve its data handling practices. This strategy must express both ethical principles and expected behavior related to data, expressed in values statements and a code of ethical behavior. The component pieces of such a strategy include:

- **Values statements:** Values statements describe what the organization believes in. Examples might include truth, fairness, or justice. These statements provide a framework for ethical handling of data and decision-making.
- **Ethical data handling principles:** Ethical data handling principles describe how an organization approaches challenges presented by data; for example, how to respect the right of individuals to privacy. Principles and expected behaviors can be summarized in a code of ethics and supported through an ethics policy. Socialization of the code and policy should be included in the training and communications plan.
- **Compliance framework:** A compliance framework includes factors that drive organizational obligations. Ethical behaviors should enable the organization to meet compliance requirements. Compliance requirements are influenced by geographic and sector concerns.
- **Risk assessments:** Risk assessments identify the likelihood and the implications of specific problems arising within the organization. These should be used to prioritize actions related to mitigation, including employee compliance with ethical principles.
- **Training and communications:** Training should include review of the code of ethics. Employee must sign off that they are familiar with the code and the implications of unethical handling of data. Training needs to be ongoing; for example, through a requirement for an annual ethics statement affirmation. Communications should reach all employees.
- **Roadmap:** The roadmap should include a timeline with activities that can be approved by management. Activities will include execution of the training and communications plan, identification and remediation of gaps in existing practices, risk mitigation, and monitoring plans. Develop detailed statements that reflect the target position of the organization on the appropriate handling of data, include roles, responsibilities, and processes, and references to experts for more information. The roadmap should cover all applicable laws, and cultural factors.

- **Approach to auditing and monitoring:** Ethical ideas and the code of ethics can be reinforced through training. It is also advisable to monitor specific activities to ensure that they are being executed in compliance with ethical principles.

3.5.4 Adopt a Socially Responsible Ethical Risk Model

Data professionals involved in Business Intelligence, analytics, and Data Science are often responsible for data that describes:

- Who people are, including their countries of origin and their racial, ethnic, and religious characteristics
- What people do, including political, social, and potentially criminal activities
- Where people live, how much money they have, what they buy, who they talk with or text or send email to
- How people are treated, including outcomes of analysis, such as scoring and preference tracking that will tag them as ultimately privileged or not for future business

This data can be misused and counteract the principles underlying data ethics: respect for persons, beneficence, and justice.

Executing BI, analytics, and Data Science activities fairly requires an ethical perspective that looks beyond the boundaries of the organization for which people work and accounts for implications to the wider community. An ethical perspective is necessary not only because data can easily be misused but also because organizations have a social responsibility not to do harm with their data.

For example, an organization might set criteria for what it considers ‘bad’ customers in order to stop doing business with those individuals. But if that organization has a monopoly on an essential service in a particular geographic area, then some of those individuals may find themselves without that essential service and they will be in harm’s way because of the organization’s decision.

Projects that use personal data should have a disciplined approach to the use of that data. See Figure 13. They should account for:

- How they select their populations for study (arrow 1)
- How data will be captured (arrow 2)
- What activities analytics will focus on (arrow 3)
- How the results will be made accessible (arrow 4)

Within each area of consideration, they should address potential ethical risks, with a particular focus on possible negative effects on customers or citizens.

A risk model can be used to determine whether to execute the project. It will also influence how to execute the project. For example, the data will be made anonymous, the private information removed from the file, the security on the files tightened or confirmed, and a review of the local and other applicable privacy law reviewed with legal. Dropping customers may not be permitted under law if the organization is a monopoly in a jurisdiction, and citizens have no other provider options such as energy or water.

Because data analytics projects are complex, people may not see the ethical challenges. Organizations need to actively identify potential risks. They also need to protect whistleblowers who do see risks and raise concerns. Automated monitoring is not sufficient protection from unethical activities. People – the analysts themselves – need to reflect on possible bias. Cultural norms and ethics in the workplace influence corporate behavior – learn and use the ethical risk model. DAMA International encourages data professionals to take a professional stand, and present the risk situation to business leaders who may not have recognized the implications of particular uses of data and these implications in their work.

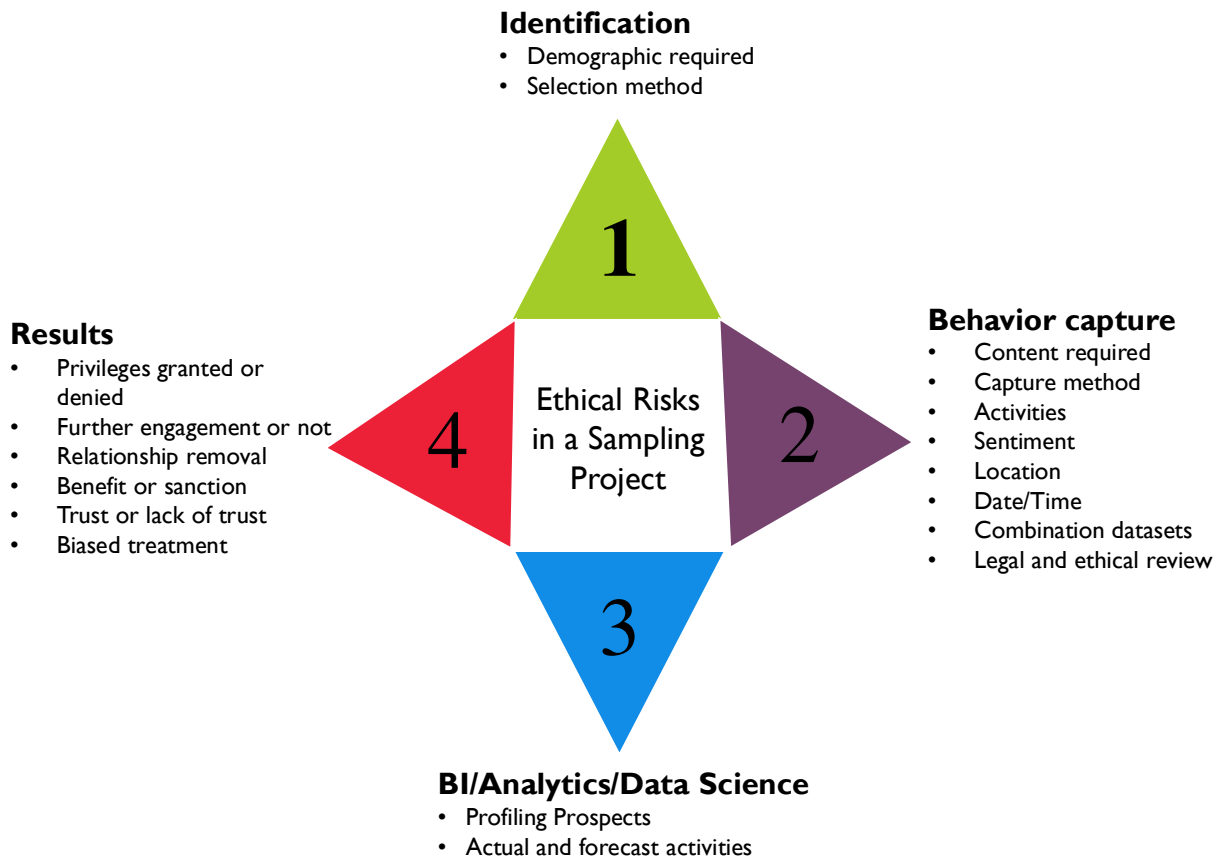


Figure 13 Ethical Risk Model for Sampling Projects

3.6 Data Ethics and Governance

Oversight for the appropriate handling of data falls under both data governance and legal counsel. Together they are required to keep up-to-date on legal changes, and reduce the risk of ethical impropriety by ensuring employees are aware of their obligations. Data Governance must set standards and policies for and provide oversight of data handling practices. Employees must expect fair handling, protection from reporting possible breaches, and non-interference in their personal lives. Data Governance has a particular oversight requirement to review plans and decisions proposed by BI, analytics and Data Science studies.

DAMA International's Certified Data Management Professional (CDMP) certification requires that data management professional subscribe to a formal code of ethics, including an obligation to handle data ethically for the sake of society beyond the organization that employs them.

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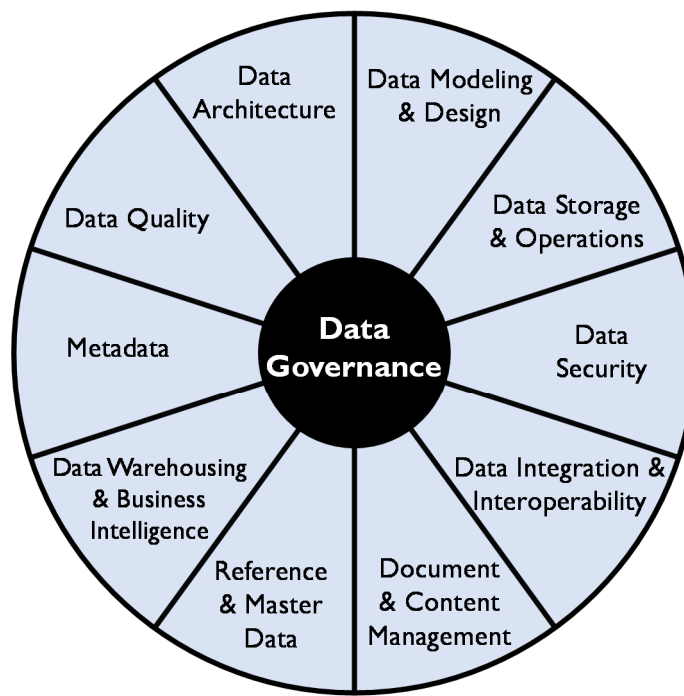
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Data Governance



DAMA-DMBOK2 Data Management Framework

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1. Introduction

Data Governance (DG) is defined as the exercise of authority and control (planning, monitoring, and enforcement) over the management of data assets. All organizations make decisions about data, regardless of whether they have a formal Data Governance function. Those that establish a formal Data Governance program exercise authority and control with greater intentionality (Seiner, 2014). Such organizations are better able to increase the value they get from their data assets.

The Data Governance function guides all other data management functions. The purpose of Data Governance is to ensure that data is managed properly, according to policies and best practices (Ladley, 2012). While the driver of data management overall is to ensure an organization gets value out of its data, Data Governance focuses on how

decisions are made about data and how people and processes are expected to behave in relation to data. The scope and focus of a particular data governance program will depend on organizational needs, but most programs include:

- **Strategy:** Defining, communicating, and driving execution of Data Strategy and Data Governance Strategy
- **Policy:** Setting and enforcing policies related to data and Metadata management, access, usage, security, and quality
- **Standards and quality:** Setting and enforcing Data Quality and Data Architecture standards
- **Oversight:** Providing hands-on observation, audit, and correction in key areas of quality, policy, and data management (often referred to as *stewardship*)
- **Compliance:** Ensuring the organization can meet data-related regulatory compliance requirements
- **Issue management:** Identifying, defining, escalating, and resolving issues related to data security, data access, data quality, regulatory compliance, data ownership, policy, standards, terminology, or data governance procedures
- **Data management projects:** Sponsoring efforts to improve data management practices
- **Data asset valuation:** Setting standards and processes to consistently define the business value of data assets

To accomplish these goals, a Data Governance program will develop policies and procedures, cultivate data stewardship practices at multiple levels within the organization, and engage in organizational change management efforts that actively communicate to the organization the benefits of improved data governance and the behaviors necessary to successfully manage data as an asset.

For most organizations, adopting formal Data Governance requires the support of organizational change management (see Chapter 17), as well as sponsorship from a C-level executive, such as Chief Risk Officer, Chief Financial Officer, or Chief Data Officer.

The ability to create and share data and information has transformed our personal and economic interactions. Dynamic market conditions and a heightened awareness of data as a competitive differentiator are causing organizations to realign data management responsibilities. This type of change is clear in the financial, ecommerce, government, and retail sectors. Organizations increasingly strive to become data-driven – proactively considering data requirements as part of strategy development, program planning, and technology implementation. However, doing so often entails significant cultural challenges. Moreover, because culture can derail any strategy, Data Governance efforts need to include a cultural change component – again, supported by strong leadership.

To benefit from data as a corporate asset, the organizational *culture* must learn to value data and data management activities. Even with the best data strategy, data governance and data management plans will not succeed unless the organization accepts and manages change. For many organizations, cultural change is a major challenge. One of the foundational tenets of change management is that organizational change requires individual change (Hiatt and

Creasey, 2012). When data governance and data management demand significant behavioral changes, formal change management is required for success.

Data Governance and Stewardship



Figure 14 Context Diagram: Data Governance and Stewardship

1.1 Business Drivers

The most common driver for data governance is often regulatory compliance, especially for heavily regulated industries, such as financial services and healthcare. Responding to evolving legislation requires strict data governance processes. The explosion in advanced analytics and Data Science has created an additional driving force.

While compliance or analytics may drive governance, many organizations back into data governance via an information management program driven by other business needs, such as Master Data Management (MDM), by major data problems, or both. A typical scenario: a company needs better customer data, it chooses to develop Customer MDM, and then it realizes successful MDM requires data governance.

Data governance is not an end in itself. It needs to align directly with organizational strategy. The more clearly it helps solve organizational problems, the more likely people will change behaviors and adopt governance practices. Drivers for data governance most often focus on reducing risks or improving processes.

- **Reducing Risk**
 - **General risk management:** Oversight of the risks data poses to finances or reputation, including response to legal (E-Discovery) and regulatory issues.
 - **Data security:** Protection of data assets through controls for the availability, usability, integrity, consistency, auditability and security of data.
 - **Privacy:** Control of private / confidential / Personal Identifying Information (PII) through policy and compliance monitoring.
- **Improving Processes**
 - **Regulatory compliance:** The ability to respond efficiently and consistently to regulatory requirements.
 - **Data quality improvement:** The ability to contribute to improved business performance by making data more reliable.
 - **Metadata Management:** Establishment of a business glossary to define and locate data in the organization; ensuring the wide range of other Metadata is managed and made available to the organization.
 - **Efficiency in development projects:** SDLC improvements to address issues and opportunities in data management across the organization, including management of data-specific technical debt through governance of the data lifecycle.
 - **Vendor management:** Control of contracts dealing with data, such as cloud storage, external data purchase, sales of data as a product, and outsourcing data operations.

It is essential to clarify the particular business drivers for data governance within an organization and to align them with overall business strategy. Focusing the ‘DG organization’ often alienates leadership who perceive extra overhead without apparent benefits. Sensitivity to organizational culture is necessary to determine the right language, operating model, and roles for the program. As of the writing of the DMBOK2, the term *organization* is being replaced with terms like *operating model* or *operating framework*.

While people sometimes claim it is difficult to understand what data governance is, *governance* itself is a common concept. Rather than inventing new approaches, data management professionals can apply the concepts and

principles of other types of governance to the governance of data. A common analogy is to equate data governance to auditing and accounting. Auditors and controllers set the rules for managing financial assets. Data governance professionals set rules for managing data assets. Other areas carry out these rules.

Data governance is not a one-time thing. Governing data requires an ongoing program focused on ensuring that an organization gets value from its data and reduces risks related to data. A Data Governance team can be a virtual organization or a line organization with specific accountabilities. To be effective, the roles and activities within data governance need to be well understood. They should be built around an operating framework that functions well in the organization. A data governance program should take into account distinctive organizational and cultural issues and the specific data management challenges and opportunities within the organization. (See Chapters 1 and 16.)

Data governance is separate from IT governance. IT governance makes decisions about IT investments, the IT application portfolio, and the IT project portfolio – in other words, hardware, software, and overall technical architecture. IT governance aligns the IT strategies and investments with enterprise goals and strategies. The COBIT (Control Objectives for Information and Related Technology) framework provides standards for IT governance, but only a small portion of the COBIT framework addresses managing data and information. Some critical topics, such as Sarbanes-Oxley compliance (U.S.A.), span the concerns of corporate governance, IT governance, and data governance. In contrast, Data Governance focuses exclusively on the management of data assets and of data as an asset.

1.2 Goals and Principles

The goal of Data Governance is to enable an organization to manage data as an asset. DG provides the principles, policy, processes, framework, metrics, and oversight to manage data as an asset and to guide data management activities at all levels. To achieve this overall goal, a DG program must be:

- **Sustainable:** The DG program needs to be ‘sticky’. DG is not a project with a defined end; it is an ongoing process that requires organizational commitment. DG necessitates changes in how data is managed and used. This does not always mean massive new organizations and upheaval. It does mean managing change in a way that is sustainable beyond the initial implementation of any data governance component. Sustainable data governance depends on business leadership, sponsorship, and ownership.
- **Embedded:** DG is not an add-on process. DG activities need to be incorporated into development methods for software, use of data for analytics, management of Master Data, and risk management.
- **Measured:** DG done well has positive financial impact, but demonstrating this impact requires understanding the starting point and planning for measurable improvement.

Implementing a DG program requires commitment to change. The following principles, developed since the early 2000s, can help set a strong foundation for data governance.²⁶

²⁶ The Data Governance Institute. <http://bit.ly/1ef0tnb>.

- **Leadership and strategy:** Successful Data Governance starts with visionary and committed leadership. Data management activities are guided by a data strategy that is itself driven by the enterprise business strategy.
- **Business-driven:** Data Governance is a business program, and, as such, must govern IT decisions related to data as much as it governs business interaction with data.
- **Shared responsibility:** Across all Data Management Knowledge Areas, data governance is a shared responsibility between business data stewards and technical data management professionals.
- **Multi-layered:** Data governance occurs at both the enterprise and local levels and often at levels in between.
- **Framework-based:** Because data governance activities require coordination across functional areas, the DG program must establish an operating framework that defines accountabilities and interactions.
- **Principle-based:** Guiding principles are the foundation of DG activities, and especially of DG policy. Often, organizations develop policy without formal principles – they are trying to solve particular problems. Principles can sometimes be reverse-engineered from policy. However, it is best to articulate a core set of principles and best practices as part of policy work. Reference to principles can mitigate potential resistance. Additional guiding principles will emerge over time within an organization. Publish them in a shared internal environment along with other data governance artifacts.

1.3 Essential Concepts

Just as an auditor controls financial processes but does not actually execute financial management, data governance ensures data is properly managed without directly executing data management (see Figure 15). Data governance represents an *inherent separation of duty between oversight and execution*.

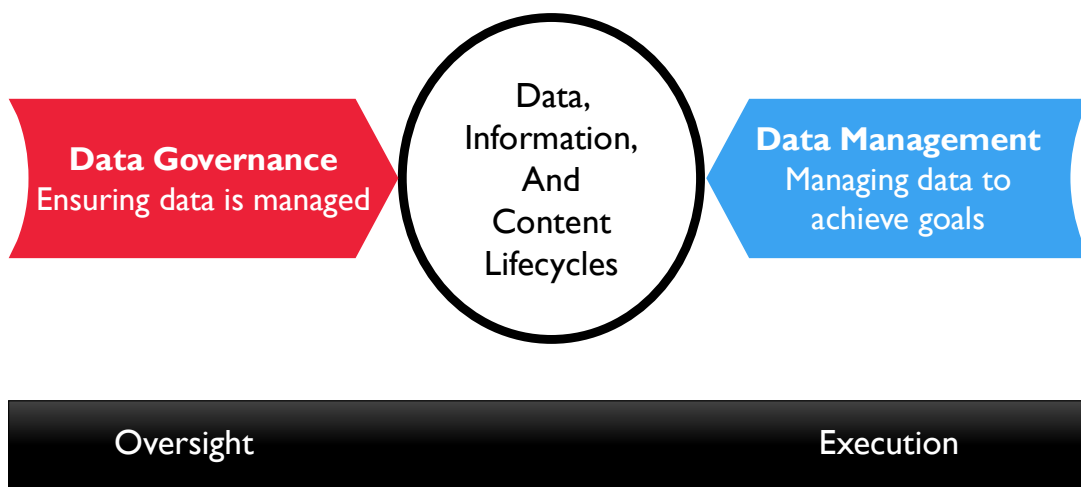


Figure 15 Data Governance and Data Management

1.3.1 Data-centric Organization

A data-centric organization values data as an asset and manages data through all phases of its lifecycle, including project development and ongoing operations. To become data-centric, an organization must change the way it translates strategy into action. Data is no longer treated as a by-product of process and applications. Ensuring data is of high quality is a goal of business processes. As organizations strive to make decisions based on insights gained from analytics, effective data management becomes a very high priority.

People tend to conflate data and information technology. To become data-centric, organizations need to think differently and recognize that managing data is different from managing IT. This shift is not easy. Existing culture, with its internal politics, ambiguity about ownership, budgetary competition, and legacy systems, can be a huge obstacle to establishing an enterprise vision of data governance and data management.

While each organization needs to evolve its own principles, those that seek to get more value from their data are likely to share the following:

- Data should be managed as a corporate asset
- Data management best practices should be incented across the organization
- Enterprise data strategy must be directly aligned with overall business strategy
- Data management processes should be continuously improved

1.3.2 Data Governance Organization

The core word in governance is *govern*. Data governance can be understood in terms of political governance. It includes legislative-like functions (defining policies, standards, and the Enterprise Data Architecture), judicial-like functions (issue management and escalation), and executive functions (protecting and serving, administrative responsibilities). To better manage risk, most organizations adopt a representative form of data governance, so that all stakeholders can be heard.

Each organization should adopt a governance model that supports its business strategy and is likely to succeed within its own cultural context. Organizations should also be prepared to evolve that model to meet new challenges. Models differ with respect to their organizational structure, level of formality, and approach to decision-making. Some models are centrally organized, while others are distributed.

Data governance organizations may also have multiple layers to address concerns at different levels within an enterprise – local, divisional, and enterprise-wide. The work of governance is often divided among multiple committees, each with a purpose and level of oversight different from the others.

Figure 16 represents a generic data governance model, with activities at different levels within the organization (vertical axis), as well as separation of governance responsibilities within organizational functions and between technical (IT) and business areas. Table 4 describes the typical committees that might be established within a data governance operating framework. Note this is not an organization chart. The diagram explains how various areas work together to carry out DG, in-line with the aforementioned trend to de-emphasize the term *organization*.

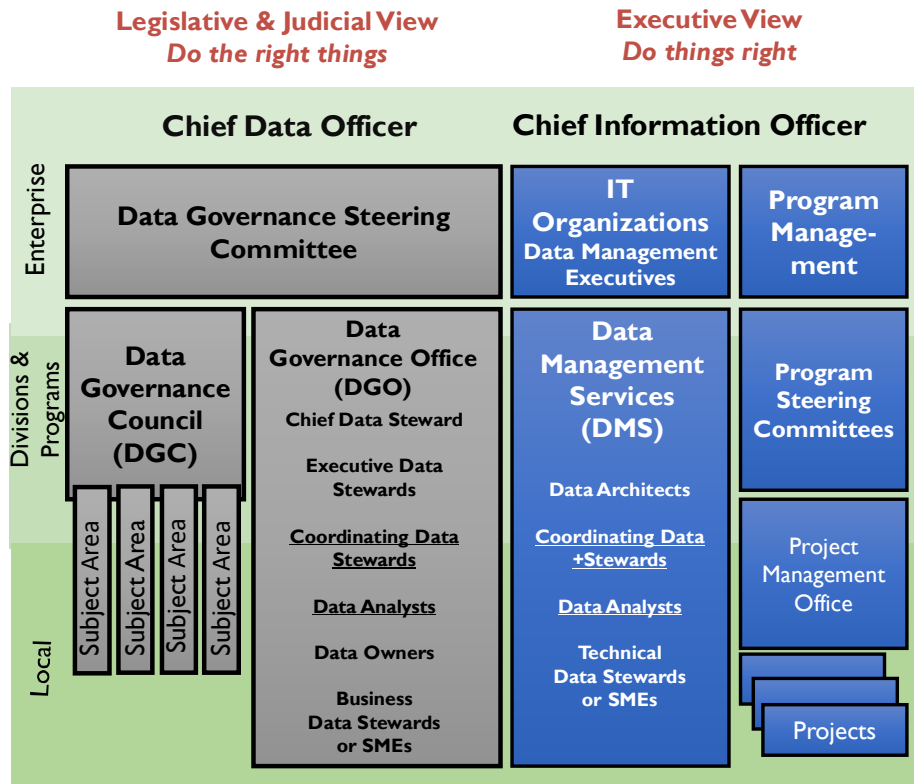


Figure 16 Data Governance Organization Parts

Table 4 Typical Data Governance Committees / Bodies

Data Governance Body	Description
Data Governance Steering Committee	The primary and highest authority organization for data governance in an organization, responsible for oversight, support, and funding of data governance activities. Consists of a cross-functional group of senior executives. Typically releases funding for data governance and data governance-sponsored activities as recommended by the DGC and CDO. This committee may in turn have oversight from higher-level funding or initiative-based steering committees.
Data Governance Council (DGC)	Manages data governance initiatives (e.g., development of policies or metrics), issues, and escalations. Consists of executive according to the operating model used. See Figure 17.
Data Governance Office (DGO)	Ongoing focus on enterprise-level data definitions and data management standards across all DAMA-DMBOK Knowledge Areas. Consists of coordinating roles that are labelled as <i>data stewards</i> or <i>custodians</i> , and <i>data owners</i> .
Data Stewardship Teams	Communities of interest focused on one or more specific subject-areas or projects, collaborating or consulting with project teams on data definitions and data management standards related to the focus. Consists of business and technical data stewards and data analysts.
Local Data Governance Committee	Large organizations may have divisional or departmental data governance councils working under the auspices of an Enterprise DGC. Smaller organizations should try to avoid such complexity.

1.3.3 Data Governance Operating Model Types

In a centralized model, one Data Governance organization oversees all activities in all subject areas. In a replicated model, the same DG operating model and standards are adopted by each business unit. In a federated model, one Data Governance organization coordinates with multiple Business Units to maintain consistent definitions and standards. (See Figure 17 and Chapter 16.)

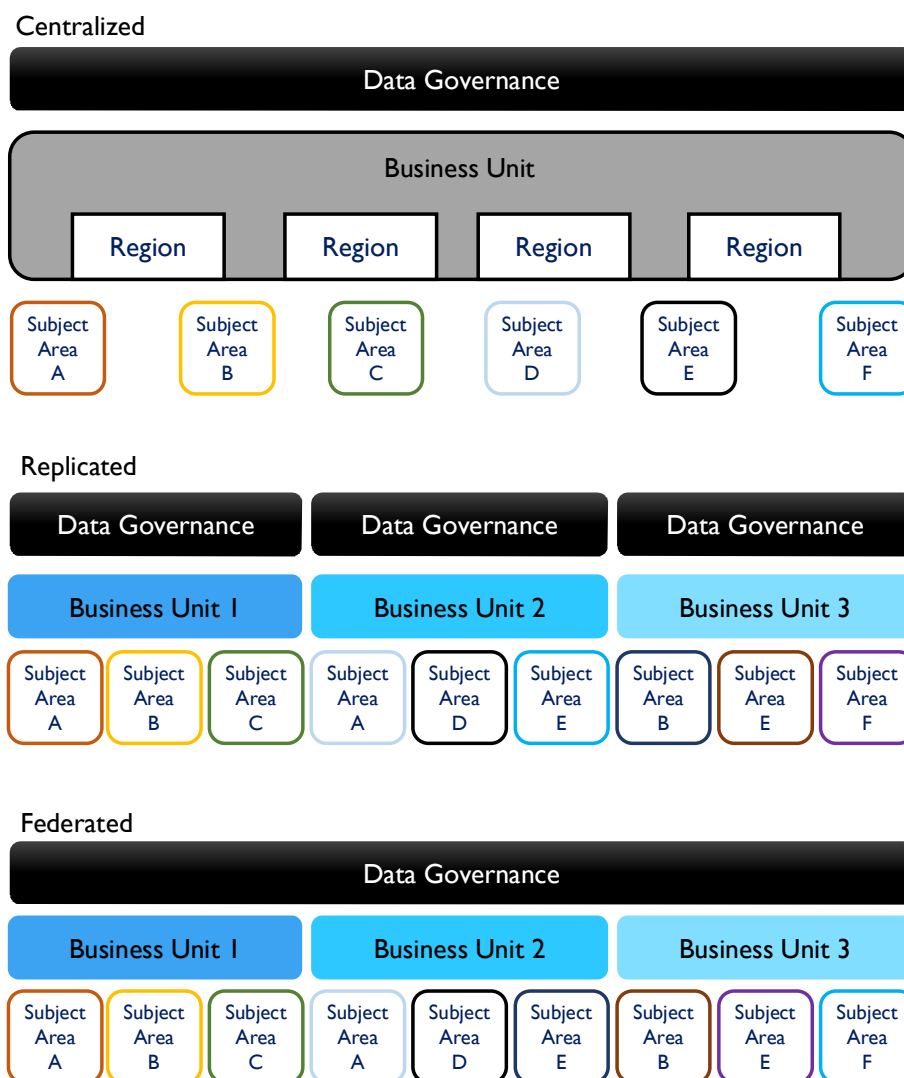


Figure 17 Enterprise DG Operating Framework Examples²⁷

1.3.4 Data Stewardship

Data Stewardship is the most common label to describe accountability and responsibility for data and processes that ensure effective control and use of data assets. Stewardship can be formalized through job titles and

²⁷ Adapted from Ladley (2012).

descriptions, or it can be a less formal function driven by people trying to help an organization get value from its data. Often terms like *custodian* or *trustee* are synonyms for those who carry out steward-like functions.

The focus of stewardship activities will differ from organization to organization, depending on organizational strategy, culture, the problems an organization is trying to solve, its level of data management maturity, and the formality of its stewardship program. However, in most cases, data stewardship activities will focus on some, if not all, of the following:

- **Creating and managing core Metadata:** Definition and management of business terminology, valid data values, and other critical Metadata. Stewards are often responsible for an organization's Business Glossary, which becomes the system of record for business terms related to data.
- **Documenting rules and standards:** Definition/documentation of business rules, data standards, and data quality rules. Expectations used to define high quality data are often formulated in terms of rules rooted in the business processes that create or consume data. Stewards help surface these rules in order to ensure that there is consensus about them within the organization and that they are used consistently.
- **Managing data quality issues:** Stewards are often involved with the identification and resolution of data related issues or in facilitating the process of resolution.
- **Executing operational data governance activities:** Stewards are responsible for ensuring that, day-to-day and project-by-project, data governance policies and initiatives are adhered to. They should influence decisions to ensure that data is managed in ways that support the overall goals of the organization.

1.3.5 Types of Data Stewards

A *steward* is a person whose job it is to manage the property of another person. Data Stewards manage data assets on behalf of others and in the best interests of the organization (McGilvray, 2008). Data Stewards represent the interests of all stakeholders and must take an enterprise perspective to ensure enterprise data is of high quality and can be used effectively. Effective Data Stewards are accountable and responsible for data governance activities and have a portion of their time dedicate to these activities.

Depending on the complexity of the organization and the goals of its DG program, formally appointed Data Stewards may be differentiated by their place within an organization, by the focus of their work, or by both. For example:

- **Chief Data Stewards** may chair data governance bodies in lieu of the CDO or may act as a CDO in a virtual (committee-based) or distributed data governance organization. They may also be Executive Sponsors.
- **Executive Data Stewards** are senior managers who serve on a Data Governance Council.
- **Enterprise Data Stewards** have oversight of a data domain across business functions.

- **Business Data Stewards** are business professionals, most often recognized subject matter experts, accountable for a subset of data. They work with stakeholders to define and control data.
- **A Data Owner** is a business Data Steward, who has approval authority for decisions about data within their domain.
- **Technical Data Stewards** are IT professionals operating within one of the Knowledge Areas, such as Data Integration Specialists, Database Administrators, Business Intelligence Specialists, Data Quality Analysts or Metadata Administrators.
- **Coordinating Data Stewards** lead and represent teams of business and technical Data Stewards in discussions across teams and with executive Data Stewards. Coordinating Data Stewards are particularly important in large organizations.

The first edition of the DAMA-DMBOK stated that “the best Data Stewards are often found, not made” (DAMA, 2009). This assertion acknowledges that in most organizations, there are people who steward data, even in the absence of a formal data governance program. Such individuals are already involved in helping the organization reduce data-related risks and get more value from its data. Formalizing their stewardship accountabilities recognizes the work they are doing and enables them to be more successful and to contribute more. All of that said, Data Stewards can be ‘made’; people can be trained to be Data Stewards. And people who are already stewarding data can develop their skills and knowledge so that they become better at the work of stewardship (Plotkin, 2014).

1.3.6 Data Policies

Data policies are directives that codify principles and management intent into fundamental rules governing the creation, acquisition, integrity, security, quality, and use of data and information.

Data policies are global. They support data standards, as well as expected behaviors related to key aspects of data management and use. Data policies vary widely across organizations. Data policies describe the ‘what’ of data governance (what to do and what not to do), while standards and procedures describe ‘how’ to do data governance. There should be relatively few data policies, and they should be stated briefly and directly.

1.3.7 Data Asset Valuation

Data asset valuation is the process of understanding and calculating the economic value of data to an organization. Because data, information, and even Business Intelligence are abstract concepts, people have difficulty aligning them with economic impact. The key to understanding the value of a non-fungible item (like data) is understanding how it is used and the value brought by its usage (Redman, 1996). Unlike many other assets (e.g., money, physical equipment), data sets are not interchangeable (fungible). One organization’s customer data differs from another organization’s in important ways; not only the customers themselves, but the data associated with them (purchasing history, preferences, etc.) How an organization gets value from customer data (i.e., what it learns about its customers from this data and how it applies what it learns) can be a competitive differentiator.

Most phases of the data lifecycle involve costs (including acquiring, storing, administering, and disposing of data). Data only brings value when it is used. When used, data also creates costs related to risk management. So value comes when the economic benefit of using data outweighs the costs of acquiring and storing it, as well as managing risk related to usage.

Some other ways to measure value include:

- **Replacement cost:** The replacement or recovery cost of data lost in a disaster or data breach, including the transactions, domains, catalogs, documents and metrics within an organization.
- **Market value:** The value as a business asset at the time of a merger or acquisition.
- **Identified opportunities:** The value of income that can be gained from opportunities identified in the data (in Business Intelligence), by using the data for transactions, or by selling the data.
- **Selling data:** Some organizations package data as a product or sell insights gained from their data.
- **Risk cost:** A valuation based on potential penalties, remediation costs, and litigation expenses, derived from legal or regulatory risk from:
 - The absence of data that is required to be present.
 - The presence of data that should not be present (e.g., unexpected data found during legal discovery; data that is required to be purged but has not been purged).
 - Data that is incorrect, causing damage to customers, company finances, and reputation in addition to the above costs.
 - Reduction in risk and risk cost is offset by the operational intervention costs to improve and certify data

To describe the concept of information asset value, one can translate Generally Accepted Accounting Principles into Generally Accepted Information Principles²⁸ (see Table 5).

Table 5 Principles for Data Asset Accounting

Principle	Description
Accountability Principle	An organization must identify individuals who are ultimately accountable for data and content of all types.
Asset Principle	Data and content of all types are assets and have characteristics of other assets. They should be managed, secured, and accounted for as other material or financial assets.
Audit Principle	The accuracy of data and content is subject to periodic audit by an independent body.
Due Diligence Principle	If a risk is known, it must be reported. If a risk is possible, it must be confirmed. Data risks include risks related to poor data management practices.
Going Concern Principle	Data and content are critical to successful, ongoing business operations and management (i.e., they are not viewed as temporary means to achieve results or merely as a business by-product).

²⁸ Adapted from Ladley (2010). See pp 108-09, Generally Accepted Information Principles.

Principle	Description
Level of Valuation Principle	Value the data as an asset at a level that makes the most sense, or is the easiest to measure.
Liability Principle	There is a financial liability connected to data or content based on regulatory and ethical misuse or mismanagement.
Quality Principle	The meaning, accuracy, and lifecycle of data and content can affect the financial status of the organization.
Risk Principle	There is risk associated with data and content. This risk must be formally recognized, either as a liability or through incurring costs to manage and reduce the inherent risk.
Value Principle	There is value in data and content, based on the ways these are used to meet an organization's objectives, their intrinsic marketability, and/or their contribution to the organization's goodwill (balance sheet) valuation. The value of information reflects its contribution to the organization offset by the cost of maintenance and movement.

2. Activities

2.1 Define Data Governance for the Organization

Data Governance efforts must support business strategy and goals. An organization's business strategy and goals inform both the enterprise data strategy and how data governance and data management activities need to be operationalized in the organization.

Data governance enables shared responsibility for data-related decisions. Data governance activities cross organizational and system boundaries in support of an integrated view of data. Successful data governance requires a clear understanding of what is being governed and who is being governed, as well as who is governing.

Data governance is most effective when it is an enterprise effort, rather than isolated to a particular functional area. Defining the scope of data governance in an enterprise usually entails defining what *enterprise* means. Data governance, in turn, governs that defined enterprise.

2.2 Perform Readiness Assessment

Assessments that describe the current state of an organization's information management capabilities, maturity, and effectiveness are crucial to planning a DG program. Because they can be used to measure a program's effectiveness, assessments are also valuable in managing and sustaining a DG program.

Typical assessments include:

- **Data management maturity:** Understand what the organization does with data; measure its current data management capabilities and capacity. The focus is on the impressions business personnel have about how well the company manages data and uses data to its advantage, as well as on objective criteria, such as use of tools, levels of reporting, etc. (See Chapter 15.)
- **Capacity to change:** Since DG requires behavioral change, it is important to measure the capacity for the organization to change behaviors required for adapting DG. Secondly, this activity will help identify potential resistance points. Often DG requires formal organizational change management. In assessing the capacity to change, the change management process will evaluate existing organizational structure, perceptions of culture, and the change management process itself (Hiatt and Creasey, 2012). (See Chapter 17.)
- **Collaborative readiness:** This assessment characterizes the organization's ability to collaborate in the management and use of data. Since stewardship by definition crosses functional areas, it is collaborative in nature. If an organization does not know how to collaborate, culture will be an obstacle to stewardship. Never assume an organization knows how to collaborate. When done in conjunction with change capacity this assessment offers insight into the cultural capacity for implementing DG.
- **Business alignment:** Sometimes included with the change capacity, a business alignment assessment examines how well the organization aligns uses of data with business strategy. It is often surprising to discover how ad hoc data-related activities can be.

2.3 Perform Discovery and Business Alignment

A DG program must contribute to the organization by identifying and delivering on specific benefits (e.g., reduce fines paid to regulators). Discovery activity will identify and assess the effectiveness of existing policies and guidelines – what risks they address, what behaviors they encourage, and how well they have been implemented. Discovery can also identify opportunities for DG to improve the usefulness of data and content. Business alignment attaches business benefits to DG program elements.

Data Quality (DQ) analysis is part of discovery. DQ assessment will provide insight into existing issues and obstacles, as well as the impact and risks associated with poor quality data. DQ assessment can identify business processes that are at risk if executed using poor quality data, as well as the financial and other benefits of creating a Data Quality program as part of data governance efforts. (See Chapter 13.)

Assessment of data management practices is another key aspect of the data governance discovery process. For example, this might mean identifying power users to create an initial list of potential agents for ongoing DG activity.

Derive a list of DG requirements from the discovery and alignment activities. For example, if regulatory risks generate a financial concern to the business, then specify DG activities that support risk management. These requirements will drive DG strategy and tactics.

2.4 Develop Organizational Touch Points

Part of alignment includes developing organizational touchpoints for Data Governance work. Figure 18 illustrates examples of touch points that support alignment and cohesiveness of an enterprise data governance and data management approach in areas outside the direct authority of the Chief Data Officer.

- **Procurement and Contracts:** The CDO works with Vendor/Partner Management or Procurement to develop and enforce standard contract language vis-à-vis data management contracts. These could include Data-as-a-Service (DaaS) and cloud-related procurements, other outsourcing arrangements, third-party development efforts, or content acquisition/ licensing deals, and possibly data-centric IT tools acquisitions and upgrades.
- **Budget and Funding:** If the CDO is not directly in control of all data acquisition-related budgets, then the office can be a focal point for preventing duplicate efforts and ensuring optimization of acquired data assets.
- **Regulatory Compliance:** The CDO understands and works within required local, national, and international regulatory environments, and how these impact the organization and their data management activities. Ongoing monitoring is performed to identify and track new and potential impacts and requirements.
- **SDLC / development framework:** The data governance program identifies control points where enterprise policies, processes, and standards can be developed in the system or application development lifecycles.

The touch points that the CDO influences support the organization's cohesiveness in managing its data, therefore, increasing its nimbleness to use its data. In essence, this is a vision of how DG will be perceived by the organization.

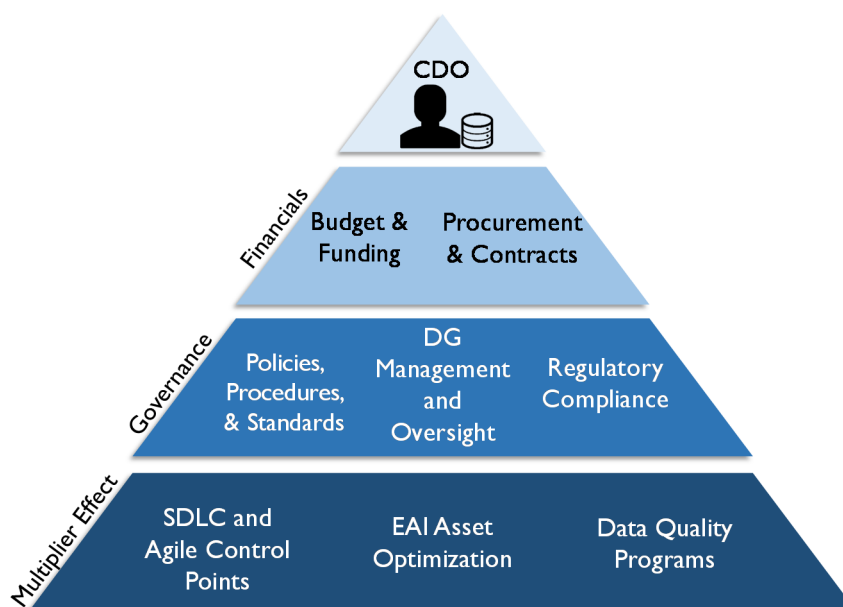


Figure 18 CDO Organizational Touch Points

2.5 Develop Data Governance Strategy

A data governance strategy defines the scope and approach to governance efforts. DG strategy should be defined comprehensively and articulated in relation to the overall business strategy, as well as to data management and IT strategies. It should be implemented iteratively as the pieces are developed and approved. The specific content will be tailored to each organization, but the deliverables include:

- **Charter:** Identifies the business drivers, vision, mission, and principles for data governance, including readiness assessment, internal process discovery, and current issues or success criteria
- **Operating framework and accountabilities:** Defines structure and responsibility for data governance activities
- **Implementation roadmap:** Timeframes for the roll out of policies and directives, business glossary, architecture, asset valuation, standards and procedures, expected changes to business and technology processes, and deliverables to support auditing activities and regulatory compliance
- **Plan for operational success:** Describing a target state of sustainable data governance activities

2.6 Define the DG Operating Framework

While developing a basic definition of DG is easy, creating an operating model that an organization will adopt can be difficult. Consider these areas when constructing an organization's operating model:

- **Value of data to the organization:** If an organization sells data, obviously DG has a huge business impact. Organizations that use data as a crucial commodity (e.g., Facebook, Amazon) will need an operating model that reflects the role of data. For organizations where data is an operational lubricant, the form of DG will be less intense.
- **Business model:** Decentralized business vs. centralized, local vs. international, etc. are factors that influence how business occurs, and therefore, how the DG operating model is defined. Links with specific IT strategy, Data Architecture, and application integration functions should be reflected in the target operating framework design (per Figure 16).
- **Cultural factors:** Such as acceptance of discipline and adaptability to change. Some organizations will resist the imposition of governance by policy and principle. Governance strategy will need to advocate for an operating model that fits with organizational culture, while still progressing change.
- **Impact of regulation:** Highly regulated organizations will have a different mindset and operating model of DG than those less regulated. There may be links to the Risk Management group or Legal as well.

Layers of data governance are often part of the solution. This means determining where accountability should reside for stewardship activities, who owns the data, etc. The operating model also defines the interaction between the governance organization and the people responsible for data management projects or initiatives, the engagement of change management activities to introduce this new program, and the model for issue management resolution

pathways through governance. Figure 19 shows an example of an operating framework. The example is illustrative. This kind of artifact must be customized to meet the needs of a specific organization.

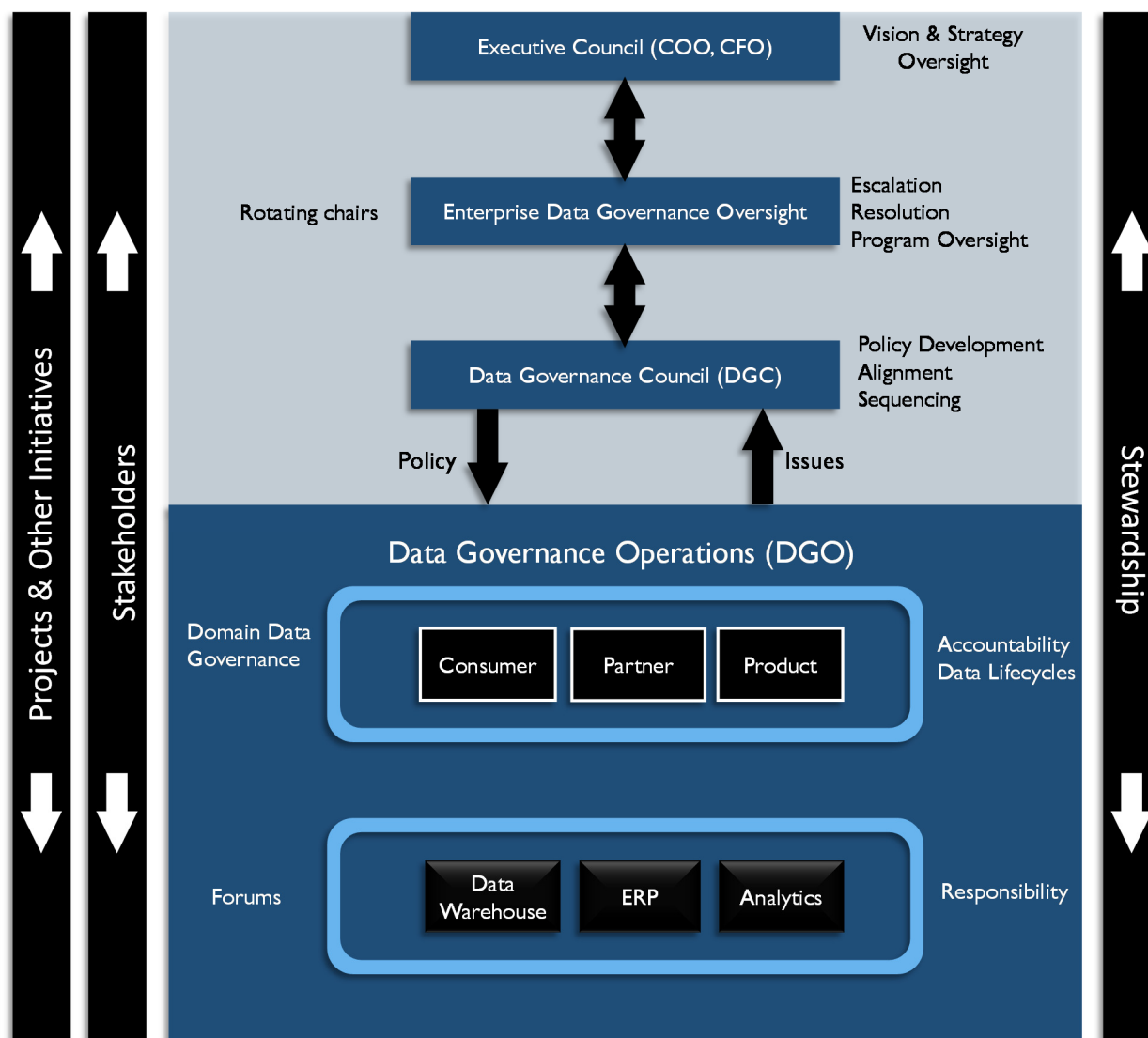


Figure 19 An Example of an Operating Framework

2.7 Develop Goals, Principles, and Policies

Development of goals, principles, and policies derived from the Data Governance Strategy will guide the organization into the desired future state.

Goals, principles, and policies are typically drafted by either by data management professionals, business policy staff, or a combination, under the auspices of data governance. Next, Data Stewards and management review and refine them. Then, the Data Governance Council (or similar body) conducts the final review, revision, and adoption.

Policies may take different shapes, as in the following examples:

- The Data Governance Office (DGO) will certify data for use by the organization.
- Business owners will be approved by the Data Governance Office.
- Business owners will designate Data Stewards from their business capability areas. The Data Stewards will have day-to-day responsibility for coordinating data governance activities.
- Whenever possible, standardized reporting and/or dashboards/scorecards will be made available to serve the majority of business needs.
- Certified Users will be granted access to Certified Data for ad hoc /non-standard reporting.
- All certified data will be evaluated on a regular basis to assess its accuracy, completeness, consistency, accessibility, uniqueness, compliance, and efficiency.

Data policies must be effectively communicated, monitored, enforced, and periodically re-evaluated. The Data Governance Council may delegate this authority to the Data Stewardship Steering Committee.

2.8 Underwrite Data Management Projects

Initiatives to improve data management capabilities provide enterprise-wide benefits. These usually require cross-functional sponsorship or visibility from the DGC. They can be hard to sell because they can be perceived as obstacles to ‘just getting things done’. The key to promoting them is to articulate the ways they improve efficiency and reduce risk. Organizations that want to get more value from their data need to prioritize development of or improvement of data management capabilities.

The DGC helps define the business case and oversees project status and progress on data management improvement projects. The DGC coordinates its efforts with a Project Management Office (PMO), where one exists. Data management projects may be considered part of the overall IT project portfolio.

The DGC may also coordinate data management improvement efforts with large programs with enterprise-wide scope. Master Data Management projects, such as Enterprise Resource Planning (ERP), Customer or Citizen Relationship Management (CRM), global parts lists, are good candidates for this kind of coordination.

Data management activity in other projects must be accommodated by the internal SDLC, service delivery management, other Information Technology Infrastructure Library (ITIL) components, and PMO processes.²⁹ Every project with a significant data component (and almost every project has these) should capture data management requirements *early* in the SDLC (planning and design phases). These include architecture, regulatory compliance, system-of-record identification and analysis, and data quality inspection and remediation. There may also be data management support activities, including requirements verification testing using standard test beds.

²⁹ <http://bit.ly/2spRr7e>.

2.9 Engage Change Management

Organizational Change Management (OCM) is the vehicle for bringing about change in an organization's systems and processes. The Change Management Institute posits that organizational change management is more than just the 'people side of projects'. It should be viewed as the approach the whole organization uses to manage change well. Organizations often manage the transitions of projects rather than the evolution of the organization (Anderson and Ackerson, 2012). An organization that is mature in its management of change builds a clear organizational vision, actively leads and monitors change from the top, and designs and manages smaller change efforts. It adapts change initiatives based on the feedback and collaboration of the whole organization (Change Management Institute, 2012). (See Chapter 17.)

For many organizations, the formality and discipline inherent in DG differ from existing practices. Adopting them requires that people change their behaviors and interactions. A formal OCM program, with the right executive sponsor, is critical to driving the behavioral changes required to sustain DG. Organizations should create a team responsible for:

- **Planning:** Planning change management, including performing stakeholder analysis, gaining sponsorship, and establishing a communications approach to overcome resistance to change.
- **Training:** Creating and executing training plans for data governance programs.
- **Influencing systems development:** Engaging with the PMO to add data governance steps the SDLC.
- **Policy implementation:** Communicating data policies and the organization's commitment to data management activities.
- **Communications:** Increasing awareness of the role and responsibilities of Data Stewards and other data governance professionals, as well as the objectives and expectations for data management projects.

Communications are vital to the change management process. A change management program supporting formal Data Governance should focus communications on:

- **Promoting the value of data assets:** Educate and inform employees about the role data plays in achieving organizational goals.
- **Monitoring and acting on feedback about data governance activities:** In addition to sharing information, communications plans should elicit feedback that can guide both the DG program and the change management process. Actively seeking and using input from stakeholders can build commitment to the program's goals, while also identifying successes and opportunities for improvement.
- **Implementing data management training:** Training at all levels of the organization increases awareness of data management best practices and processes.
- Measuring the effects of change management on in five key areas:³⁰

³⁰ <http://bit.ly/1qKvLyJ>. See also Hiatt and Creasey (2012).

- Awareness of the need to change
 - Desire to participate and support the change
 - Knowledge about how to change
 - Ability to implement new skills and behaviors
 - Reinforcement to keep the change in place
- **Implementing new metrics and KPIs:** Employee incentives should be realigned to support behaviors connected to data management best practices. Since enterprise data governance requires cross-functional cooperation, incentives should encourage cross-unit activities and collaboration.

2.10 Engage in Issue Management

Issue management is the process for identifying, quantifying, prioritizing, and resolving data governance-related issues, including:

- **Authority:** Questions regarding decision rights and procedures
- **Change management escalations:** Issues arising from the change management process
- **Compliance:** Issues with meeting compliance requirements
- **Conflicts:** Conflicting policies, procedures, business rules, names, definitions, standards, architecture, data ownerships and conflicting stakeholder interests in data and information
- **Conformance:** Issue related to conformance to policies, standards, architecture, and procedures
- **Contracts:** Negotiation and review of data sharing agreements, buying and selling data, and cloud storage
- **Data security and identity:** Privacy and confidentiality issues, including breach investigations
- **Data quality:** Detection and resolution of data quality issues, including disasters or security breaches

Many issues can be resolved locally in Data Stewardship teams. Issues requiring communication and / or escalation must be logged, and may be escalated to the Data Stewardship teams, or higher to the DGC, as shown in Figure 20. A Data Governance Scorecard can be used to identify trends related to issues, such as where within the organization they occur, what their root causes are, etc. Issues that cannot be resolved by the DGC should be escalated to corporate governance and / or management.

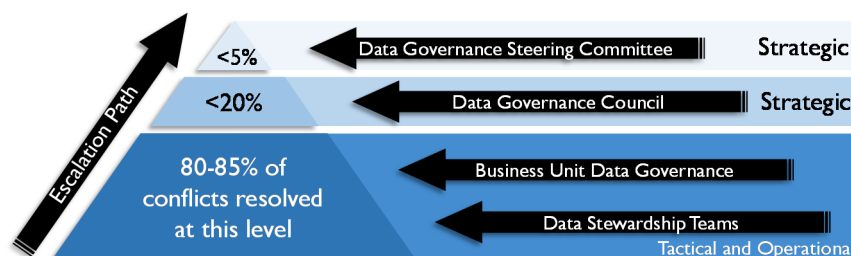


Figure 20 Data Issue Escalation Path

Data governance requires control mechanisms and procedures for:

- Identifying, capturing, logging, tracking, and updating issues
- Assignment and tracking of action items
- Documenting stakeholder viewpoints and resolution alternatives
- Determining, documenting, and communicating issue resolutions
- Facilitating objective, neutral discussions where all viewpoints are heard
- Escalating issues to higher levels of authority

Data issue management is very important. It builds credibility for the DG team, has direct, positive effects on data consumers, and relieves the burden on production support teams. Solving issues also proves that data can be managed and its quality improved. Successful issue management requires control mechanisms that demonstrate the work effort and impact of resolution.

2.11 Assess Regulatory Compliance Requirements

Every enterprise is affected by governmental and industry regulations, including regulations that dictate how data and information are to be managed. Part of the data governance function is to monitor and ensure regulatory compliance. Regulatory compliance is often the initial reason for implementing data governance. Data governance guides the implementation of adequate controls to monitor and document compliance with data-related regulations.

Several global regulations have significant implications on data management practices. For example:

- **Accounting Standards:** The Government Accounting Standards Board (GASB) and the Financial Accounting Standards Board (FASB) accounting standards also have significant implications on how information assets are managed (in the US).
- **BCBS 239** (Basel Committee on Banking Supervision) and **Basel II** refer to Principles for Effective Risk Data Aggregation and risk reporting, a wide ranging set of regulations for banks. Since 2006, financial institutions doing business in European Union countries are required to report standard information proving liquidity.
- **CPG 235:** The Australian Prudential Regulation Authority (APRA) provides oversight of banking and insurance entities. It publishes standards and guides to assist in meeting these standards. Among these is CGP 235, a standard for managing data risk. It focuses on addressing the sources of data risk and on managing data throughout its lifecycle.
- **PCI-DSS:** The Payment Card Industry Data Security Standards (PCI-DSS).
- **Solvency II:** European Union regulations, similar to Basel II, for the insurance industry.
- **Privacy laws:** Local, sovereign, and international laws all apply.

Data governance organizations work with other business and technical leadership to evaluate the implications of regulations. The organization must determine, for example,

- In what ways is a regulation relevant to the organization?
- What constitutes compliance? What policies and procedures will be required to achieve compliance?
- When is compliance required? How and when is compliance monitored?
- Can the organization adopt industry standards to achieve compliance?
- How is compliance demonstrated?
- What is the risk of and penalty for non-compliance?
- How is non-compliance identified and reported? How is non-compliance managed and rectified?

DG monitors the organization's response to regulatory requirements or audit undertakings involving data and data practices (for example, certifying the quality of data in regulatory reporting). (See Chapter 6.)

2.12 Implement Data Governance

Data governance cannot be implemented overnight. It requires planning – not only to account for organizational change, but also simply because it includes many complex activities that need to be coordinated. It is best to create an implementation roadmap that illustrates the timeframes for and relationship between different activities. For example, if the DG program is focused on improving compliance, priorities may be driven by specific regulatory requirements. In a federated DG organization, implementation in various lines of business can occur on different schedules, based on their level of engagement and maturity, as well as funding.

Some DG work is foundational. Other work depends on it. This work has an initial release and ongoing cultivation. Prioritized activities in the early stages include:

- Defining data governance procedures required to meet high priority goals
- Establishing a business glossary and documenting terminology and standards
- Coordinating with Enterprise Architecture and Data Architecture to support better understanding of the data and the systems
- Assigning financial value to data assets to enable better decision-making and to increase understanding of the role that data plays in organizational success

2.13 Sponsor Data Standards and Procedures

A standard is defined as “something that is very good and that is used to make judgments about the quality of other things” or as “something set up and established by authority as a rule for the measure of quantity, weight, extent, value, or quality.”³¹ Standards help define quality because they provide a means of comparison. They also offer the potential to simplify processes. By adopting a standard, an organization makes a decision once and codifies it in a

³¹ <http://bit.ly/2sTfugb>

set of assertions (the standard). It does not need to make the same decision all over again for each project. Enforcing standards should promote consistent results from the processes using them.

Unfortunately, creating or adopting standards is often a politicized process and these goals get lost. Most organizations are not well-practiced at developing or enforcing data or data governance standards. In some cases, they have not recognized the value in doing so and therefore have not taken the time to do so. Other times they simply don't know how to. Consequently, 'standards' vary widely within and across organizations, as do expectations for conformance. DG standards should be mandatory.

Data standards can take different forms depending on what they describe: assertions about how a field must be populated, rules governing the relationships between fields, detailed documentation of acceptable and unacceptable values, format, etc. They are usually drafted by data management professionals. Data standards should be reviewed, approved and adopted by the DGC, or a delegated workgroup, such as a Data Standards Steering Committee. The level of detail in data standards documentation depends, in part, on organizational culture. Keep in mind that documenting data standards presents an opportunity to capture details and knowledge that otherwise may be lost. Recreating or reverse engineering to access this knowledge is very expensive, compared to documenting it up front.

Data standards must be effectively communicated, monitored, and periodically reviewed and updated. Most importantly, there must be a means to enforce them. Data can be measured against standards. Data management activities can be audited for standards compliance by the DGC or the Data Standards Steering Committee on a defined schedule or as part of SDLC approval processes.

Data management procedures are the documented methods, techniques, and steps followed to accomplish specific activities that produce certain outcomes and supporting artifacts. Like policies and standards, procedures vary widely across organizations. As is the case with data standards, procedural documents capture organizational knowledge in an explicit form. Procedural documentation is usually drafted by data management professionals.

Examples of concepts that can be standardized within the Data Management Knowledge Areas include:

- **Data Architecture:** Enterprise data models, tool standards, and system naming conventions
- **Data Modeling and Design:** Data model management procedures, data modeling naming conventions, definition standards, standard domains, and standard abbreviations
- **Data Storage and Operations:** Tool standards, standards for database recovery and business continuity, database performance, data retention, and external data acquisition
- **Data Security:** Data access security standards, monitoring and audit procedures, storage security standards, and training requirements
- **Data Integration:** Standard methods and tools used for data integration and interoperability
- **Documents and Content:** Content management standards and procedures, including use of enterprise taxonomies, support for legal discovery, document and email retention periods, electronic signatures, and report distribution approaches

- **Reference and Master Data:** Reference Data Management control procedures, systems of data record, assertions establishing and mandating use, standards for entity resolution
- **Data Warehousing and Business Intelligence:** Tool standard, processing standards and procedures, report and visualization formatting standards, standards for Big Data handling
- **Metadata:** Standard business and technical Metadata to be captured, Metadata integration procedures and usage
- **Data Quality:** Data quality rules, standard measurement methodologies, data remediation standards and procedures
- **Big Data and Data Science:** Data source identification, authority, acquisition, system of record, sharing and refresh

2.14 Develop a Business Glossary

Data Stewards are generally responsible for business glossary content. A glossary is necessary because people use words differently. It is particularly important to have clear definitions for data, because data represents things other than itself (Chisholm, 2010). In addition, many organizations develop their own internal vocabulary. A glossary is a means of sharing this vocabulary within the organization. Developing and documenting standard data definitions reduces ambiguity and improves communication. Definitions must be clear, rigorous in wording, and explain any exceptions, synonyms or variants. Approvers of terminology should include representatives from core user groups. Data Architecture often can supply draft definitions and type breakouts from subject area models.

Business glossaries have the following objectives:

- Enable common understanding of the core business concepts and terminology
- Reduce the risk that data will be misused due to inconsistent understanding of the business concepts
- Improve the alignment between technology assets (with their technical naming conventions) and the business organization
- Maximize search capability and enable access to documented institutional knowledge

A business glossary is not merely a list of terms and definitions. Each term will also be associated with other valuable Metadata: synonyms, metrics, lineage, business rules, the steward responsible for the term, etc.

2.15 Coordinate with Architecture Groups

The DGC sponsors and approves data architecture artifacts, such as a business-oriented enterprise data model. The DGC may appoint or interact with an Enterprise Data Architecture Steering Committee or Architecture Review Board (ARB) to oversee the program and its iterative projects. The enterprise data model should be developed and

maintained jointly by data architects and Data Stewards working together in subject area teams. Depending on the organization, this work can be coordinated either by the Enterprise Data Architect or by the steward. As business requirements evolve, the Data Stewardship teams should propose changes and develop extensions to the enterprise data model.

The enterprise data model should be reviewed, approved, and formally adopted by the DGC. This model must align with key business strategies, processes, organizations, and systems. Data strategy and Data Architecture are central to coordination between the ‘Doing things right’ and ‘Doing the right things’ when managing data assets.

2.16 Sponsor Data Asset Valuation

Data and information are assets because they have or can create value. Today’s accounting practices consider data an intangible asset, much like software, documentation, expert knowledge, trade secrets, and other intellectual property. That said, organizations find it challenging to put monetary value on data. The DGC should organize the effort and set standards for doing so.

Some organizations start by estimating the value of business losses due to inadequate information. Information gaps – the difference between what information is needed and what is available – represent business liabilities. The cost of closing or preventing gaps can be used to estimate of business value of the missing data. From there, the organization can develop models to estimate the value of the information that does exist.

Value estimates can be built into a data strategy roadmap that will justify business cases for root cause solutions to quality issues, as well as for other governance initiatives.

2.17 Embed Data Governance

One goal of the data governance organization is to embed in a range of processes behaviors related to managing data as an asset. The ongoing operation of DG requires planning. The operations plan contains the list of events required to implement and operate DG activities. It outlines activities, timing, and techniques necessary to sustain success.

Sustainability means acting to ensure that processes and funding are in place to enable the continued performance of the DG organizational framework. Central to this requirement is that the organization *accepts* the governance of data; that the function is managed, its results are monitored and measured, and the obstacles that so often cause DG programs to falter or fail are overcome.

In order to deepen the organization’s understanding of data governance in general, its application locally, and to learn from each other, create a Data Governance Community of Interest. This is particularly useful in the first years of governance, and will likely taper off as the DG operations become mature.

3. Tools and Techniques

Data governance is fundamentally about organizational behavior. This is not a problem that can be solved through technology. However, there are tools that support the overall process. For example, DG requires ongoing communication. A DG program should take advantage of existing communications channels to communicate key messages in a consistent manner and to keep stakeholders informed about policies, standards, and requirements.

In addition, a DG program must manage its own work and its own data effectively. Tools help not only with these tasks, but also with the metrics that support them. Before choosing a tool for a specific function, like a business glossary solution, an organization should define its overall governance goals and requirements with an eye to building out a tool set. For example, some glossary solutions include additional components for policy and workflow management. If such additional functionality is desired, requirements should be clarified and tested before a tool is adopted. Otherwise, the organization will have multiple tools, none of which may meet its needs.

3.1 Online Presence / Websites

The Data Governance program should have an online presence. It can make core documents available via a central website or a collaboration portal. Websites can house documentation libraries, give access to search capabilities, and help manage simple workflow. A website can also help establish a brand for the program through logos and a consistent visual representation. A DG program website should include:

- The Data Governance strategy and program charter, including vision, benefits, goals, principles, and implementation roadmap
- Data policies and data standards
- Descriptions of data stewardship roles and responsibilities
- Program news announcements
- Links to forums for a Data Governance Community of Interest
- Links to executive messages regarding data governance topics
- Reports on Data Quality measurements
- Procedures for issue identification and escalation
- Links to request services or capture issues
- Documents, presentations, and training programs with links to related online resources
- Data Governance program contact information

3.2 Business Glossary

A Business Glossary is a core DG tool. It houses agreed-upon definitions of business terms and relates these to data. There are many business glossary tools available, some as part of larger ERP systems, data integration tools, or Metadata management tools, and some as standalone tools.

3.3 Workflow Tools

Larger organizations may want to consider a robust workflow tool to manage processes, such as the implementation of new data governance policies. These tools connect processes to documents, and can be useful in policy administration and issue resolution.

3.4 Document Management Tools

Very often, a document management tool is used by governance teams to assist in managing policies and procedures.

3.5 Data Governance Scorecards

The collection of metrics to track data governance activities and compliance with policies can be reported up to the Data Governance Council and Data Governance Steering Committees on an automated scorecard.

4. Implementation Guidelines

Once the data governance program is defined, an operating plan developed, and an implementation roadmap prepared with the support of information gathered in the data maturity assessment (see Chapter 15), the organization can begin to implement processes and policies. Most rollout strategies are incremental, either applying DG first to a large effort, such as MDM, or by a region or division. Rarely is DG deployed enterprise-wide as a first effort.

4.1 Organization and Culture

As noted in Section 2.9, the formality and discipline inherent in data governance will be new and different for many organizations. Data governance adds value by bringing about changes in behavior. There may be resistance to change and a learning or adoption curve for new methods of making decisions and governing projects.

Effective and long-lasting data governance programs require a cultural shift in organizational thinking and behavior about data, as well as an ongoing program of change management to support the new thinking, behaviors, policies, and processes to achieve the desired future state of behavior around data. No matter how precise or exotic the data governance strategy, ignoring culture will diminish chances for success. Focus on managing change must be part of the implementation strategy.

The target of organization change is sustainability. *Sustainability* is a quality of a process that measures how easy it is for the process to continue to add value. Sustaining a data governance program requires planning for change. (See Chapter 17.)

4.2 Adjustment and Communication

Data Governance programs are implemented incrementally within the context of a wider business and data management strategy. Success requires keeping the wider goals in mind while putting the pieces in place. The DG team will need to be flexible and adjust its approach as conditions shift. Tools required to manage and communicate changes include:

- **Business / DG strategy map:** This map connects DG activity with business needs. Periodically measuring and communicating how DG is helping the business is vital to obtain ongoing support for the program.
 - **DG roadmap:** The roadmap to DG should not be rigid. It should be adapted to changes in business environment or priorities.
 - **Ongoing business case for DG:** The business case must be adjusted periodically to reflect changing priorities and financial realities of the organization.
 - **DG metrics:** Metrics will need to grow and change as the DG program matures.
-

5. Metrics

To counter resistance or the challenge of a long learning curve, a DG program must be able to measure progress and success through metrics that demonstrate how DG participants have added business value and attained objectives.

In order to manage the behavior changes required, it is important to measure progress of the rollout of data governance, compliance with the data governance requirements, and the value data governance is bringing to the organization. Metrics that reinforce DG's value and those that verify the organization has the resources required to support DG after it is rolled out are also important to sustaining a DG program. Sample metrics include:

- Value
 - Contributions to business objectives
 - Reduction of risk
 - Improved efficiency in operations
- Effectiveness
 - Achievement of goals and objectives
 - Extent stewards are using the relevant tools
 - Effectiveness of communication
 - Effectiveness of education/training

- Speed of change adoption
- Sustainability
 - Performance of policies and processes (i.e., are they working appropriately?)
 - Conformance to standards and procedures (i.e., are staff following the guidance and changing behavior as necessary?)

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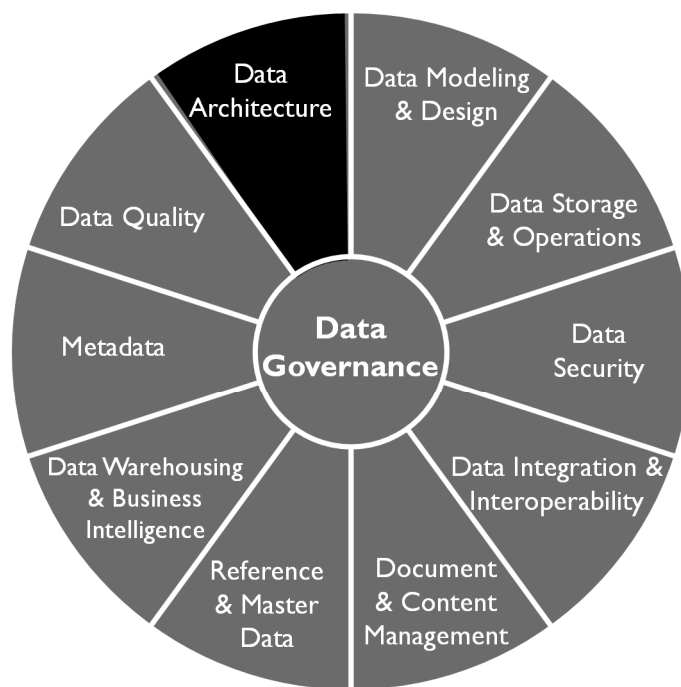
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Data Architecture



DAMA-DMBOK2 Data Management Framework

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1. Introduction

Architecture refers to the art and science of building things (especially habitable structures) and to the results of the process of building – the buildings themselves. In a more general sense, architecture refers to an organized arrangement of component elements intended to optimize the function, performance, feasibility, cost, and aesthetics of an overall structure or system.

The term *architecture* has been adopted to describe several facets of information systems design. ISO/IEC 42010:2007 *Systems and Software Engineering – Architecture Description* (2011) defines *architecture* as “the

fundamental organization of a system, embodied in its components, their relationships to each other and the environment, and the principles governing its design and evolution.” However, depending on context, the word *architecture* can refer to a description of the current state of systems, the components of a set of systems, the discipline of designing systems (architecture practice), the intentional design of a system or a set of systems (future state or proposed architecture), the artifacts that describe a system (architecture documentation), or the team that does the design work (the Architects or the Architecture team).

Architecture practice is carried out at different levels within an organization (enterprise, domain, project, etc.) and with different areas of focus (infrastructure, application, and data). Exactly what architects do can be confusing to people who are not architects and who do not recognize the distinctions implied by these levels and focus areas. One reason architectural frameworks are valuable is that they enable non-architects to understand these relationships.

The discipline of Enterprise Architecture encompasses domain architectures, including business, data, application, and technology. Well-managed enterprise architecture practices help organizations understand the current state of their systems, promote desirable change toward future state, enable regulatory compliance, and improve effectiveness. Effective management of data and the systems in which data is stored and used is a common goal of the breadth of architecture disciplines.

In this chapter, Data Architecture will be considered from the following perspectives:

- **Data Architecture outcomes**, such models, definitions and data flows on various levels, usually referred as Data Architecture artifacts
- **Data Architecture activities**, to form, deploy and fulfill Data Architecture intentions
- **Data Architecture behavior**, such as collaborations, mindsets, and skills among the various roles that affect the enterprise’s Data Architecture

Together, these three form the essential components of Data Architecture.

Data Architecture is fundamental to data management. Because most organizations have more data than individual people can comprehend, it is necessary to represent organizational data at different levels of abstraction so that it can be understood and management can make decisions about it.

Data Architecture artifacts includes specifications used to describe existing state, define data requirements, guide data integration, and control data assets as put forth in a data strategy. An organization’s Data Architecture is described by an integrated collection of master design documents at different levels of abstraction, including standards that govern how data is collected, stored, arranged, used, and removed. It is also classified by descriptions of all the containers and paths that data takes through an organization’s systems.

The most detailed Data Architecture design document is a formal enterprise data model, containing data names, comprehensive data and Metadata definitions, conceptual and logical entities and relationships, and business rules. Physical data models are included, but as a product of data modeling and design, rather than Data Architecture.